

BIOLOGY GRADE 11: GROWTH AND DEVELOPMENT IN PLANTS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.2 (10 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	11
Subject	Biology
Strand	Strand 2.0: Anatomy and Physiology of Plants
Sub-Strand	Sub-Strand 2.2: Growth and Development in Plants
Total Duration	10 lessons × 40 minutes = 400 minutes total
Sub-Strand Content	– Meaning and significance of growth and development – Structure of the seed (monocots and dicots) – Seed dormancy (meaning, significance, causes and ways of breaking seed dormancy) – Germination (process, types and conditions necessary for germination) – Primary and secondary growth – Measurement of growth and growth curves – Role of hormones in plant growth and development (include apical dominance)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - distinguish between growth and development in living organisms, - relate structures of monocot and dicot seeds to their functions, - outline causes and ways of breaking dormancy in seeds, - investigate conditions necessary for seed germination, - investigate types of germination in plants, - illustrate primary and secondary growth in flowering plants, - demonstrate the role of hormones in plant growth and development, - appreciate the importance of growth and development in plants.
Core Competencies	– Citizenship: The learner solves problems in the community by sharing the seedlings from the established tree nursery. – Learning to Learn: The learner reflects on their own work and adjusts accordingly while carrying out activities to demonstrate ways of breaking seed dormancy. – Critical Thinking and Problem Solving: Learners investigate the farmer's problem of uneven germination and growth of maize and beans. – Communication and Collaboration: Learners share evidence and findings in group investigations. – Creativity and Imagination: Learners design experiments and build and revise models. – Digital Literacy: Learners use the maize and bean time-lapse videos and other digital resources, with teacher-prepared alternatives where there is no connectivity.
Core Values	– Unity: The learner works harmoniously with others while carrying out experiments to investigate conditions necessary for seed

	germination. – Patriotism: The learner demonstrates devotion to the country/community while sharing seedlings with the school community for environmental conservation, and links healthy plant growth to Kenya's food security and agricultural development. – Responsibility: Learners care for the experimental seeds and follow procedures. – Integrity: Learners record actual observations rather than changing results to match predictions. – Respect: Learners listen to and consider other learners' explanations.
Science & Engineering Practices	– Asking Questions: Learners write questions after the time-lapse videos and post them on the Driving Question Board. – Developing and Using Models: Learners draw initial models of maize and bean growth and revise them after each lesson, including models of secondary growth. – Planning and Carrying Out Investigations: Learners test conditions for germination, types of germination and regions of growth using maize and bean seeds. – Analyzing and Interpreting Data: Learners record seedling height or stem width and draw growth curves. – Constructing Explanations: Learners explain differences in germination, emergence and growth using evidence on seed structure, dormancy, conditions and hormones. – Engaging in Argument from Evidence: Learners discuss and defend explanations with peers. – Obtaining, Evaluating and Communicating Information: Learners use readings, videos and animations on hormones and growth.
Pertinent & Contemporary Issues (PCIs)	– Financial Literacy: The learner explores ways of generating income while selling seedlings to members of the community. – Prevention of Global Warming: The learner mitigates the effects of global warming by establishing a tree nursery and distributing seedlings in the community. – Food security and nutrition: Understanding germination and growth supports better maize and bean production. – Climate change and adaptation: Rainfall and temperature affect germination and crop establishment. – Sustainable agriculture, environmental conservation and water conservation: Use of resources in crop and seedling production. – Technology and innovation: Use of hormones and time-lapse technology in agriculture.
Career Connections	– Agronomist: How can farmers improve crop establishment and growth? – Seed Technologist: Why do some seeds germinate while others fail to germinate? – Agricultural Research Scientist: What conditions produce better germination and plant growth? – Plant Physiologist: What processes and hormones control plant growth? – Plant Breeder: Can varieties be developed that germinate and grow better under Kenyan conditions? – Agricultural Extension Officer: How can this knowledge be transferred to farmers?
Focus for Lessons	This unit uses the puzzle of uneven germination, emergence and growth in a farmer's maize and beans to drive learning about seed structure, dormancy, germination, primary and secondary growth, growth measurement and hormones. Learners start with time-lapse videos, predict, investigate with locally available seeds in small groups, and revise a cumulative model each lesson. The unit ends in a tree nursery project in which stem cuttings are rooted using natural hormone sources such as bulb onions, Aloe and cinnamon.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: What causes the differences in germination, emergence and growth of maize and bean plants? KICD KEY INQUIRY QUESTIONS: 1. How does growth occur in plants?

Anchoring Phenomenon	A farmer plants maize and beans on the same day after rainfall. Some seeds germinate within a few days, while others take longer. Some seeds fail to germinate even though they appear healthy. As the plants grow, the stems become taller and thicker, roots extend deeper, and some plants develop branches. Learners first watch time-lapse videos of a bean seedling and a maize seedling germinating and growing, so they see the process before explaining it. It is puzzling because the seeds were planted on the same day, in the same field, with the same rain, yet they behave very differently. Healthy-looking seeds do not germinate, maize and beans emerge differently, and growth is not just getting taller since stems also thicken and branch. What explains these differences?
Storyline Thread	Lesson 1: The learner gets to know the phenomena. Watch the time-lapse videos of maize and beans (Bean Time-Lapse - 25 days Soil cross section; Growing corn time lapse), draw initial models, and write questions to post on the Driving Question Board. Question raised: What is growth and development? Lesson 2: Meaning of growth and development. Structure of the seed (monocot and dicot). Discuss the meaning of growth and development, observe, draw and label monocot and dicot seeds, and discuss the functions of the structures with peers. Question raised: What is seed dormancy? Lesson 3: Seed dormancy. Examine seed specimens to identify causes of seed dormancy and discuss with peers; carry out activities to demonstrate ways of breaking seed dormancy. Question raised: How does germination occur? Lesson 4: Germination (process, types and conditions necessary for germination). Use maize and bean seeds to investigate conditions necessary for germination and carry out experiments to investigate types of germination. Question raised: How do plants show growth? Lesson 5: Primary and secondary growth. Observe stem sections under a microscope, draw and label them, and use drawings and models to illustrate stages of secondary growth. Question raised: How can you measure growth in plants? Lesson 6: Measurement of growth and growth curves. Use seedlings to determine regions of growth, record data, draw growth curves and discuss regions of growth with peers. Question raised: Do we have hormones in plants? Lesson 7: Role of hormones in plant growth and development. Watch a video on the effect of plant hormones, read about the roles and functions of plant hormones (including apical dominance), and revise the model to explain the differences in germination, emergence and growth of maize and beans. Lesson 8: Apply the hormone ideas to the tree nursery project by rooting stem cuttings with natural hormone sources (bulb onions, Aloe, cinnamon). Lesson 9: Consolidate the evidence from all lessons, finalise the cumulative model and explanation for the farmer's maize and beans, and review the Driving Question Board. Lesson 10: Present the final models and explanations, reflect on learning, and plan sharing and selling of seedlings to the community for conservation and income.
Supporting Phenomena	– Time-lapse video: Bean Time-Lapse - 25 days Soil cross section, showing bean germination and root and shoot growth below and above the soil. – Time-lapse video: Growing corn time lapse (#greentimelapse #gtl #timelapse), showing maize seedling germination and growth.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.

LESSON 1 (80 minutes (double lesson); can be split into 2 x 40 minutes): Meeting the Phenomenon: Mr. Wafula's Maize and Beans

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the unit by letting learners meet the phenomenon: maize H614 and bean Rosecoco planted on the same day, in the same field, after the same rain on Mr. Wafula's farm near Kitale, yet they germinate, emerge and grow differently. Learners observe, describe, draw their first models and post their questions on the Driving Question Board.
Knowledge	Learners describe differences in germination, emergence and growth between maize and beans using the time-lapse events and Day 14 counts: Maize radicle Day 4, emergence Day 7, first leaf Day 10; Bean radicle Day 3, emergence Day 8, first true leaves Day 12, branching by Day 25. Maize 88 of 100 and beans 80 of 100 germinated by Day 14. Maize is taller from Day 14 onward. Learners recognise that growth is more than getting taller (stems also thicken and branch) and that healthy-looking seeds can fail to germinate.
Skills	Observing and recording from a time-lapse video; comparing two sets of data; drawing and labelling an initial model; asking testable scientific questions; communicating ideas in simple, clear scientific English (SEP: Developing and Using Models; Asking Questions; Analyzing and Interpreting Data).
Attitudes	Curiosity about how crops grown by Kenyan farmers develop; willingness to share tentative ideas without fear of being wrong (Unity); respect for peers' ideas. Core competency: Learning to Learn, where the learner reflects on their own work and adjusts accordingly. PCI link: Financial Literacy, since a farmer's seed losses affect income.
Key Inquiry Question	How does growth occur in plants? (Driving question: What causes the differences in germination, emergence and growth of maize and bean plants?)
Purpose in Storyline	This is the opening lesson. It builds on no earlier lesson. It sets the puzzle, records learners' baseline ideas and opens the DQB. The question it raises for Lesson 2 is: What is growth and development? Lesson 2 will then teach definitions and seed structure, so they are not taught here.
Safety Notes	Handle seeds with clean hands; some commercial maize seed is treated with fungicide and must not be put in the mouth, so wash hands after handling. Keep walkways clear when learners move to the DQB. If the class is large, use groups of 5 to 6 and share one screen or the teacher's printed stills. If no electricity or internet is available, use the teacher-prepared still-image strips and the events table described below.

B. LESSON OVERVIEW

Madam Achieng opens with a story: Mr. Wafula, a farmer near Kitale in Trans-Nzoia, planted maize H614 and Rosecoco beans on the same day (Day 0), 3 cm deep, after 25 mm of rain. Yet some seeds came up quickly, some slowly and some never germinated even though they looked healthy. Learners first predict and sketch their initial ideas, then watch the two time-lapse videos (Bean Time-Lapse - 25 days | Soil cross section; Growing corn time lapse) and record what they see in an events table. They are shown the Day 14 germination counts (maize 88/100, beans 80/100) and the height trend (maize taller from Day 14 onward). Using Think-Pair-Share they draw an initial model and complete sentence starters. The teacher does not mark these as right or wrong. In groups, learners write questions on sticky notes or cards and post them on the Driving Question Board, grouping them into categories. The lesson ends with an exit ticket and the class question for Lesson 2: What is growth and development? If there is no power, the teacher uses a prepared strip of 6 labelled sketches per crop (Day 0, 3/4, 7/8, 10/12, 17/18, 25) and reads the events table aloud.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: growth development plants germination  HTML: Seed Dormancy and Germination - Advanced (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners listen to the farmer's problem: Mr. Wafula near Kitale planted maize H614 and Rosecoco beans on the same day, 3 cm deep, after 25 mm of rain, in the same field. Individually (2 min) they draw what they think happens to one maize seed and one bean seed after planting, showing root and shoot, and complete: 'I think maize and bean plants grow differently because...' and 'I think a seed needs _____ in order to germinate because...'. They then share with a partner (3 min) and a few pairs share with the class (Think-Pair-Share). The drawings are kept in their books as Day 1 baseline models.	Read the story slowly and show a real maize grain and a dry bean seed (or pictures). Say: 'Same day, same field, same rain. What do you think will happen to these seeds?' Give 2 minutes of silent individual WAIT TIME, then say: 'Turn to your partner and compare your drawings.' Circulate and listen for ideas such as 'beans need more water' or 'maize is stronger'; note them without correcting. Cold-call 4 learners (not volunteers) to share a partner's idea, saying: 'Wanjiru, tell us what your partner drew.' After each answer give 10 to 15 seconds of WAIT TIME before the next. Say clearly: 'I will not mark these right or wrong. These are your first ideas and we will return to them at the end of the unit.' Clarify words such as germinate (seed starts to grow) in simple English.	Think-Pair-Share with Predict-Observe-Explain (the 'Predict' step): committing to a prediction in writing before seeing evidence makes learners' prior ideas visible to themselves and to the teacher, so later evidence can confirm or challenge them. Cold-calling the partner's idea builds listening and safe sharing.	Observation checklist: tick learners who (a) draw both seeds, (b) show root and shoot, (c) add at least 2 labels or arrows, (d) complete both sentence starters. Look for common baseline ideas or misconceptions (e.g. 'all healthy seeds will germinate', 'growth means only getting taller', 'beans and maize germinate in the same way') and record them on a class list. No marks are given; this is a baseline.
Observe Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: growth development plants germination  HTML: Seed Dormancy and Germination - Advanced (Flexbook) <i>Related topic: not an exact match</i>	Learners watch the two time-lapse videos (Bean Time-Lapse - 25 days Soil cross section; Growing corn time lapse), each played twice. In the first viewing they just watch. In the second viewing, in groups of 5 to 6, they fill a simple events table: radicle appears (germination), shoot appears above soil (emergence), first leaf, branching. Then they study the farm data: events table (Bean: radicle Day 3, emergence Day 8, first true leaves Day 12, trifoliolate leaves Day 18, branching by Day 25; Maize: radicle Day 4, emergence Day 7, first leaf Day 10, 3 leaves Day 17, seed food store stays underground), Day 14	Before playing, say: 'Watch for three things: when the root appears, when the shoot comes above the soil, and what changes besides height.' Play each video twice; pause at key moments and ask: 'What do you see happening right now?' with 10 to 15 seconds WAIT TIME. Cold-call 3 groups to report one observation each. Write the class's observations on the board in two columns, Maize and Bean. Then display or hand out the events table and Day 14 counts. Ask: 'Which crop germinated more seeds by Day 14? How many seeds did not germinate in each?' (12 and 20.) Ask: 'Which emerged first, maize	Predict-Observe-Explain (the 'Observe' step) with structured noticing: a table forces learners to compare the same events (germination, emergence, leaves) across both crops, moving them from vague impressions ('beans are slower') to specific, data-based observations. Separating 'what I notice' from 'what I wonder' keeps description apart from explanation.	Oral probing: ask 'How do you know the bean germinated before it emerged?' Listen for use of radicle versus shoot above soil. Check each group's events table: Are Day numbers correct (e.g. maize emergence Day 7, bean Day 8)? Do learners state the counts correctly (88 and 80)? Look for learners who note that growth includes thickening and branching, not only height. Record misconceptions such as confusing germination and emergence.

for this lesson. Source: CK-12 Search ARES for similar readings	germination counts out of 100 seeds (maize 88 germinated, 12 did not; beans 80 germinated, 20 did not), and the height trend (maize taller than beans from Day 14 onward). Learners complete 'I notice that...' and 'I wonder why...' statements, noting that some seeds that looked healthy did not germinate, and that stems also thicken and branch.	or beans?' (Maize, Day 7 versus Day 8, although the bean radicle came first, Day 3 versus Day 4.) Ask: 'If the rain and planting day were the same, what could be different?' and give 15 seconds WAIT TIME. Do not explain causes; say: 'Hold on to that thought; we will investigate it.' If there is no power, use the prepared sketch strips and read the events table aloud, asking learners to arrange the sketches in order.		
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: growth development plants germination  HTML: Seed Dormancy and Germination - Advanced (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Individually, learners revise their drawing into an initial model that explains the farmer's puzzle: they draw a maize seed and a bean seed in the soil, show what happens after planting, show the root and shoot emerging at different times, and show how the plant develops over time. They add arrows and labels and write a short initial explanation using sentence starters: 'I think the seeds differ because...' and 'I think some healthy-looking seeds do not germinate because...'. In pairs they then use Claim-Evidence-Reasoning in a light form: claim (e.g. 'Maize and beans grow differently'), evidence from the table or counts (e.g. 'Maize emerged Day 7, beans Day 8; maize 88, beans 80'), and a tentative reason. A few learners present their models.	Say: 'Draw what you think is happening that we cannot see inside the soil. Use arrows and labels.' Give 5 minutes of silent work and 10 to 15 seconds WAIT TIME before taking questions. Circulate and ask: 'What does this arrow show?' and 'What evidence from the video or table supports this part?' Select 3 contrasting models (not the 'best') for sharing, and cold-call 2 other learners to say one thing they like about each model and one thing they are unsure about. Emphasise: 'Models are drafts. They will change.' Display a few models on the wall. Do not supply the scientific explanation; accept ideas such as water, soil, seed type, seed age, pests and oxygen, and record them on the board as 'possible factors to test'.	Model-based reasoning with Claim-Evidence-Reasoning: drawing an initial model forces learners to turn observations into a mechanism, exposing gaps (such as what is inside the seed). Linking claims to specific evidence from the events table and counts starts the habit of evidence-based explanation that is used in every later lesson. Peer comments are a first form of peer review.	Use a model check: look for evidence that the model (a) shows both seeds, (b) shows root and shoot, (c) shows change over time, (d) has labels and arrows, (e) offers at least one reason for the differences. Collect 5 to 6 models to photocopy or keep for revision at the end of the unit. Note the most common ideas for later lessons. Peer feedback quality: do learners point to evidence rather than opinion?
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library</i>	In small groups, learners discuss: 'What questions do you have after watching the videos and considering the farmer's problem?' Each group writes at least 3 questions on sticky notes or cards (e.g. Wanjiru-style questions: 'Why did some healthy-looking seeds	Post the driving question at the top of the board or a large manila paper. Say: 'Every question is welcome. Make it a question we could investigate.' Give groups 5 minutes and 10 to 15 seconds WAIT TIME after setting the task. Walk around and push on vague	Discussion and questioning with the Driving Question Board: turning curiosity into a visible, shared set of questions gives learners ownership of the inquiry and a way to connect every later lesson back to the main question. Clustering helps learners see	Check the quality of questions: tally how many are about causes (why/how) versus simple facts. Observe participation: does each group member contribute a question? Note whether learners connect questions to evidence from the videos or data. Identify

for this activity.  Search ARES for videos Search terms: growth development plants germination  HTML: Seed Dormancy and Germination - Advanced (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	not germinate?', 'Why did maize emerge before beans?', 'Why did the bean plant branch but not the maize?', 'What is growth and development?'). They post the questions on the DQB under the driving question: What causes the differences in germination, emergence and growth of maize and bean plants? As a class they cluster similar questions (seeds, conditions, growth, hormones) and choose the first question to follow up: What is growth and development?	questions: 'Can you ask that so we could test it?' Invite 5 groups to come forward (cold-call, one at a time) to read and post one question and say where it should be clustered. Group the notes with the class under headings such as Seeds, Conditions, Growth and Branching, and Other. Point out the question appearing most often: 'Many of you asked what growth and development mean. That is where we begin in Lesson 2.' If there is no sticky note supply, use cut paper and flour paste or tape, or write on the chalkboard.	patterns and decide what to investigate first.	questions needing refinement and ask learners to reword them. Record which DQB clusters appear so later lessons can return to them.
Model Building Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: growth development plants germination  HTML: Seed Dormancy and Germination - Advanced (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners return to their Day 1 drawing and add a 'Questions I still have' box and one improvement they would like to make but cannot yet (e.g. 'I do not know what is inside the seed'). They file the model in their books as the baseline for revision, and they complete an exit ticket: 'What did we observe today?' and 'How does this help answer our driving question?' They also write one thing they now want to find out. A few learners share in a gallery-style quick walk where they look at 3 posted models and leave one question.	Say: 'Keep this model safe. You will change it when you have evidence.' Remind learners that being unsure is part of science. Hand out or have learners write exit tickets on a half-page. Circulate during the quick gallery walk and say: 'Leave one question on a model that you want its owner to think about.' Give 10 to 15 seconds WAIT TIME before closing questions. Cold-call 3 learners to read an exit ticket answer aloud. Close with: 'Today we saw that maize and beans, planted on the same day at Mr. Wafula's farm near Kitale, behaved differently. Our new question is: What is growth and development? That is how we start Lesson 2.' Set optional home task: ask a parent or a farmer how they decide when a seed has failed.	Model revision (baseline step) and Gallery Walk: saving an initial model and naming what is missing creates the benchmark for revision later in the unit, and leaving questions on peers' models starts the culture of peer review. Exit tickets link the day's evidence to the driving question.	Exit ticket checks: Can learners state at least one observation with a number or day (e.g. 'maize 88, beans 80', 'maize emerged Day 7')? Can they link it to the driving question? Quick oral probing during the gallery walk: 'What is one thing your model cannot yet explain?' Record learners who cannot yet distinguish observation from explanation for follow-up in Lesson 2. Keep the models and exit tickets for revision at the end of the unit.

D. TEACHER REFLECTION

After the lesson ask yourself: Did every learner, including quiet ones, draw an initial model and post at least one idea? Which baseline ideas (e.g. 'all healthy seeds germinate', 'growth is just height') were most common, and how will I address them without giving answers too early? Were the questions on the DQB mostly about causes, and did they include the growth, branching and germination ideas we need? Did the video or sketch strips work with the available power and internet? How well did the learners use simple, clear scientific English for germination, emergence and growth, and which terms do I need to reinforce in Lesson 2? What will I change to make sure larger groups and shared materials still allow everyone to participate?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Maize H614 and bean Rosecoco were planted on the same day at Mr. Wafula's farm near Kitale. The bean radicle came out on Day 3 and emerged on Day 8; maize radicle on Day 4 and emerged on Day 7. By Day 14, 88 of 100 maize and 80 of 100 bean seeds had germinated, so 12 and 20 did not, even though some looked healthy. Maize was taller from Day 14 onward, stems thickened, and the bean branched by Day 25.
What did I learn?	The two crops behave differently even with the same rain and planting day. Germination (radicle comes out) and emergence (shoot appears above soil) are not the same event. Growth is not only getting taller: stems also thicken and plants branch. We have many questions, such as what growth and development mean and why healthy-looking seeds do not germinate.
How does this explain the phenomenon?	Our first ideas are only tentative: we think the differences may be due to the seeds, the soil, water, age or pests, but we do not yet have evidence. Our initial models and DQB questions are our baseline, and we will revise them as we collect evidence. The next question is: What is growth and development?

LESSON 2 (40 minutes (single lesson; if double period available, extend the Observe Phase)): Growth vs Development and Inside the Seed: Maize H614 and Rosecoco Beans

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners distinguish between growth and development and relate the structures of monocot (maize H614) and dicot (bean Rosecoco) seeds to their functions, so they can begin to explain why Mr. Wafula's two crops behaved differently.
Knowledge	a) Distinguish between growth and development in living organisms. b) Relate structures of monocot and dicot seeds to their functions. Growth = permanent, irreversible increase in size/dry mass (e.g. maize stem width 6 to 14 to 22 mm; bean height 4 cm at Day 7 to 44 cm at Day 42). Development = changes in complexity and form through differentiation (e.g. radicle to roots, unifoliate to trifoliate leaves, branching). Bean (dicot): testa, micropyle, hilum, two cotyledons storing food, plumule, radicle. Maize (monocot): fruit/seed coat fused, one cotyledon (scutellum), endosperm food store, coleoptile protecting plumule, coleorhiza protecting radicle.
Skills	Observing, drawing and labelling seed specimens (NGSS SEP: Developing and Using Models); asking questions; analysing data (heights and stem widths) to classify changes as growth or development; constructing explanations using Claim-Evidence-Reasoning; communicating in simple, clear scientific English.
Attitudes	Curiosity and care when handling living material; Unity: works harmoniously with group members when sharing seed specimens and hand lenses; respect for peers' ideas during Think-Pair-Share and the Gallery Walk. Learning to Learn: reflects on own Day 1 drawing and adjusts it.
Key Inquiry Question	How does growth occur in plants? (Lesson 2 focus: What do we mean by growth and development, and how are the maize and bean seeds built to support them?)
Purpose in Storyline	Lesson 1 launched the phenomenon (Mr. Wafula's maize H614 and Rosecoco beans, planted on the same day, behaving differently) and posted the DQB question 'What is growth and development?'. This lesson answers that question and gives the seed-structure vocabulary and ideas needed to later explain dormancy and germination. It ends by raising the next question: What is seed dormancy? Why can a healthy-looking seed with a testa and an embryo fail to germinate?
Safety Notes	Soak seeds overnight so they can be cut safely. Use a blunt knife or razor blade only under teacher control (teacher or one trained learner per group cuts the seed, away from the body). Do not taste seeds; some may be treated with pesticide (e.g. the purple or pink dressing on commercial seed). Wash hands after handling seeds. Keep specimens on trays or paper; wipe up spilled water to prevent slipping.

B. LESSON OVERVIEW

Madam Achieng's Form 3 class returns to Mr. Wafula's farm near Kitale. From the Lesson 1 time-lapse, the class saw maize H614 and Rosecoco beans planted on the same day with the same 25 mm rain, yet they emerged on different days (maize Day 7, beans Day 8) and grew differently. The DQB question 'What is growth and development?' now needs an answer. Learners first predict which of several changes are growth and which are development, then observe and draw soaked maize and bean seeds (using readily available seeds, hand lenses and a blunt blade, in groups of 4 to 5), and use the Day 14 stem width and height numbers as examples of growth. They explain what each seed part does, apply it to how each seed becomes a seedling (without teaching dormancy or germination conditions, which come in Lessons 3 and 4), update the DQB and revise their Day 1 model. The lesson ends with a new question: What is seed dormancy?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: growth development plants monocot  HTML: Monocots - Advanced (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Wanjiru's DQB question is read aloud: 'What is growth and development?'. Learners individually write a one-sentence prediction for each of 6 changes from Mr. Wafula's field: (1) bean shoot height rising from 4 cm at Day 7 to 44 cm at Day 42; (2) maize stem width rising from 6 mm (Day 14) to 22 mm (Day 42); (3) bean getting first unifoliate leaves on Day 12 and trifoliate leaves on Day 18; (4) bean branching by Day 25; (5) a maize seed swelling after the 25 mm rain; (6) the radicle emerging on Day 3 (bean) and Day 4 (maize). For each they write G (growth), D (development) or Both and a reason. They then predict, by quick sketch, what is inside a maize grain and a bean seed. Think-Pair-Share: think alone 2 minutes, pair for 2 minutes, share.	Display the 6 changes on the board or on a prepared chart (no electricity needed). Say: 'Look at what Mr. Wafula saw in his field. Which of these is just getting bigger, and which is getting more complex or changing form? Write your own idea first.' WAIT TIME: 10-15 seconds of silence after the question; do not rescue. Circulate and note ideas. Cold-call 4 pairs (not volunteers): 'Tell us your G, D or Both for number 3, and why.' Record learners' words verbatim on a Predict column on the board, and do NOT confirm correct answers yet. Say: 'Hold on to your predictions; we will test them with evidence.' Ask: 'What do you think is inside this seed that lets it become a plant?'	Think-Pair-Share, with Predict-Observe-Explain (POE) as the frame: committing to a prediction first exposes learners' everyday meaning of 'growth' (just getting taller) so that evidence in the Observe and Explain phases can confront it. Pair talk builds understanding by forcing learners to justify their ideas in simple scientific English before hearing the class.	Observable indicators: learners write a reason (not just a letter) for at least 4 of 6 items; at least 4 cold-called pairs give a reason using words like 'bigger', 'thicker', 'changes shape' or 'new part'. Probing to detect misconceptions: 'Is growth only getting taller?' 'Does swelling of a seed in water count as growth?' Tally on the board how many pairs predict each item as G, D or Both. Teacher notes the misconception 'growth = height only' for follow-up.
Observe Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: growth development plants monocot  HTML: Monocots - Advanced (Flexbook) <i>Related topic: not an exact match for this lesson.</i>	Groups of 4 to 5 (one set of specimens per group; share hand lenses between groups) receive soaked maize H614 grains, soaked Rosecoco bean seeds (soaked overnight) and a reading handout/textbook page on seed structure. (1) Observe the outside of each seed with a hand lens: bean testa, hilum (scar), micropyle (tiny pore); maize grain surface with the scar and the darker area marking the embryo. (2) Under teacher guidance, one learner removes the bean testa and splits the bean into two cotyledons, finding the plumule and radicle. The maize grain is cut lengthwise along the flat side to show the	Before the lesson, soak the seeds overnight; if seeds are scarce, provide 1 set per group or use photographs/teacher-drawn charts. Model safe cutting: 'I cut away from my body, on the tray, and only the cutter handles the blade.' Circulate with the observation checklist. Ask: 'What do you see at the end of the bean where it was attached to the pod? What is the tiny hole next to it for?' WAIT TIME: 10-15 seconds after each probe. Ask: 'Where would the bean's food be stored? How is that different from maize?' Cold-call 3 groups to hold up their cut seeds and name one visible structure. If a group lacks specimens, hand	Predict-Observe-Explain (the Observe step): direct observation of the real seeds and the growth data provides the evidence that learners compare with their earlier predictions. Drawing and labelling is a Developing and Using Models practice: it makes structures visible and links each part to a job (structure-function), which is the foundation for explaining why maize and beans emerge differently.	Observation checklist (tick/cross): (a) correctly identifies testa, micropyle, hilum on the bean; (b) separates cotyledons and finds plumule and radicle; (c) identifies the scutellum, endosperm and coleoptile in maize; (d) drawing is large, has title, label lines that do not cross, and correct proportions; (e) all members take part in handling and discussing. Expect at least 80% of groups to label at least 6 parts correctly. Oral probe: 'Which part would protect the shoot tip as it pushes through soil in maize?' (coleoptile). Misconceptions to catch: 'maize has no cotyledon' or 'the endosperm is the embryo'.

Source: CK-12 Search ARES for similar readings	scutellum (one cotyledon), endosperm, coleoptile covering the plumule and coleorhiza covering the radicle. (3) Draw and label both seeds in their notebooks (size at least half a page, pencil, ruled label lines, title and magnification note). (4) Complete a table: Part Where found (bean/maize) Function (e.g. testa protects the seed; micropyle lets water in; hilum is the scar where seed was attached; cotyledons store food in the bean; scutellum absorbs food from the endosperm for the maize embryo; endosperm stores food; coleoptile protects the plumule; coleorhiza protects the radicle; plumule becomes shoot; radicle becomes root). (5) Use the given measurements as growth examples: Day 14 to Day 42 stem width (maize 6, 14, 22 mm; bean 2, 4, 6 mm) and shoot height Day 7 to 42 (maize 3 to 125 cm; bean 4 to 44 cm). Learners calculate the increase in stem width (maize +16 mm, bean +4 mm).	over a teacher-prepared labelled drawing for them to annotate. Clarify difficult words (testa = seed coat, cotyledon = seed leaf, scutellum = shield-like cotyledon of maize) and write them on the board with simple meanings. If an internet-free class cannot play the time-lapse videos, recall the Day 7 maize and Day 8 bean emergence events from the Lesson 1 table.		
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: growth development plants monocot  HTML: Monocots - Advanced (Flexbook) <i>Related topic: not an exact match for this lesson.</i>	Learners return to the predictions from the Predict Phase and write a Claim-Evidence-Reasoning (CER) statement in their notebooks: Claim: which of the 6 changes are growth and which are development. Evidence: the numbers (e.g. maize stem width 6 mm at Day 14 to 22 mm at Day 42; bean height 4 cm at Day 7 to 44 cm at Day 42; bean's leaf types changing from unifoliate on Day 12 to trifoliate on Day 18; branching by Day 25) and what they saw in the seeds. Reasoning: growth is a permanent, irreversible increase in size or dry mass (so a seed only swelling with water is not true growth, because water can be lost	Ask: 'Which of the 6 changes belongs in the growth box, which in the development box, and which in both? What is your evidence?' WAIT TIME: 10-15 seconds. Cold-call 5 learners in different groups, each for a different item. Press for reasoning: 'How do you know it is permanent? Could the swelling be reversed if the seed dried?' Introduce the conventions: 'germination is the radicle emerging; emergence is the shoot appearing above the soil.' Use sentence starters on the board: 'I claim ... because the data show ... which means ...'. Address the misconception 'growth = taller only' using the stem width data: 'Maize	Claim-Evidence-Reasoning (CER): learners must link a claim (growth or development) to specific data and to a scientific definition, which deepens understanding beyond memorising definitions. Discussion and questioning connects the seed structure to the DQB question about how maize and beans behave differently.	Check CER notebooks: strong response contains a clear claim, at least 2 numbers as evidence (such as 6 to 22 mm, or 4 to 44 cm) and reasoning using 'permanent, irreversible increase' or 'differentiation / change in form'. Look for: at least 70% of learners correctly classify the stem width as growth and branching or leaf type change as development. Peer review: swap notebooks with a partner and underline the evidence in green and the reasoning in blue. Oral probe: 'Why can't we say a dry seed swelling in water is growth?' Record misconceptions on the board.

Source: CK-12 Search ARES for similar readings	again), while development is increasing complexity and form through cell differentiation (radicle becoming root, plumule becoming shoot with leaves, branching). They then use the seed structure to explain: 'Both seeds have an embryo with a radicle and plumule, but the bean stores food in two large cotyledons and the maize in an endosperm absorbed by one scutellum.' They tie this to what they saw in Lesson 1: maize pushes up a coleoptile (Day 7) while the bean pushes up an arched hypocotyl and lifts its cotyledons (Day 8). In pairs, they state one way the seed structure may help explain a difference in emergence.	widened by 16 mm and the bean by 4 mm. Is that growth if the plant did not get taller? Yes: size increased.' Do NOT teach dormancy or germination conditions; if a learner asks why some healthy seeds did not germinate, say: 'Great question. Hold on to it; we will see what the seed structure tells us in the next lesson. Put it on the DQB.'		
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: growth development plants monocot  HTML: Monocots - Advanced (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners go to the class DQB (a chart or wall space). They (1) find Wanjiru's question 'What is growth and development?' and, in groups, write an evidence-based answer on a sticky note/paper strip (e.g. 'Growth is a permanent increase in size such as maize stem width 6 to 22 mm; development is a change in form such as bean leaves changing from unifoliate to trifoliate') and place it beside the question. (2) Write new questions raised by the seed observation (e.g. 'Why do some healthy-looking bean seeds with a complete embryo not germinate?', 'Does the testa stop water entering?', 'Why does the maize leave its scutellum underground?'). (3) Link the DQB to the driving question: 'How does this lesson help explain the differences in germination, emergence and growth of maize and beans?' The key new question that the class agrees to investigate next: What is	Say: 'Choose one thing you now understand and one thing that still puzzles you. Write each on its own strip.' WAIT TIME: 10-15 seconds before groups begin writing. Cold-call 3 groups to read their strip aloud and say where it should go on the board (answered, partly answered, new). Model sorting by pinning Wanjiru's question as 'answered in part: definitions and seed parts known'. Highlight the strips that ask about healthy seeds failing: 'These strips connect to the testa and embryo we just saw. We will investigate them in the next lesson: What is seed dormancy?' Keep a copy of the DQB in the teacher's notebook in case the chart is removed.	Discussion and questioning using the Driving Question Board: the DQB makes the class's growing understanding visible, shows learners how this lesson's evidence (definitions, seed structure) fits into the larger puzzle, and lets their own questions drive the next investigation, which builds ownership.	Indicators: every group posts at least one answer strip and one new question; at least 2 questions refer to the testa, embryo or healthy seeds failing to germinate; answers use the terms growth/development correctly with an example from Mr. Wafula's field. Oral probe: 'Which of our questions was answered today and what was the evidence?' Note learners who post only vocabulary questions (what is a hilum?) and prompt them toward 'why' or 'how' questions.

	seed dormancy?			
Model Building Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: growth development plants monocot  HTML: Monocots - Advanced (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners take out their Day 1 (Lesson 1) drawing/model of how Mr. Wafula's maize and beans grow. Working individually for 5 minutes, they revise it using a different colour pen: (1) add labelled cross-section drawings of the maize H614 grain and Rosecoco bean seed inside the soil, showing the embryo parts (radicle, plumule), food stores (cotyledons in bean; endosperm and scutellum in maize) and protective parts (testa; coleoptile and coleorrhiza); (2) add arrows showing that radicle becomes root and plumule becomes shoot; (3) add a small box with the definitions of growth and development and one example of each from the field data; (4) add a question mark where the model still cannot explain why some healthy-looking seeds did not germinate. In a short Gallery Walk (3 minutes), groups display their revised models, and visitors leave one 'star' (what is good) and one 'wish' (what to improve). The lesson closes with an exit ticket.	Say: 'Use a new colour so we can see how your thinking has changed since Day 1.' WAIT TIME: 10-15 seconds, then circulate. Ask individuals: 'Which part of your model is new today? What evidence led you to add it?' Cold-call 3 learners to explain one revision. During the Gallery Walk, instruct: 'Walk clockwise, spend 30 seconds at each model, leave one star and one wish.' Collect exit tickets: (1) 'What evidence did we gather today?' (2) 'How does this help answer our driving question about maize and beans?' (3) 'One question I still have.' Close with: 'Next time we ask: What is seed dormancy? Why would a seed that looks healthy not germinate?' Homework: bring one dry bean and one maize grain from home or the market and note differences in the outside appearance of the testa.	Model revision with a Gallery Walk: revising the Day 1 model with evidence from this lesson shows how understanding changes, and peer review sharpens scientific communication. The question mark in the model keeps the gap (healthy seeds not germinating) visible, setting up Lesson 3.	Model check: look for at least 3 new labelled structures, correct direction of arrows (radicle to root, plumule to shoot), definitions with an example, and a question mark gap. Exit ticket success criteria: names at least 2 pieces of evidence (seed observations and a growth number such as 6 to 22 mm) and links them to the driving question. Peer review: stars and wishes refer to specific parts of the model. Quick quiz option (2 questions): 'Name the part of maize that protects the shoot tip.' 'Give one example of growth and one of development from the bean.'

D. TEACHER REFLECTION

1) Did learners move beyond 'growth = getting taller' by using the stem width data? Which examples convinced them? 2) How many groups correctly identified the scutellum, endosperm and coleoptile in maize; did sharing specimens or hand lenses slow groups down? 3) Which seed-structure terms needed the most clarification in simple English? 4) Did the CER writing show evidence plus reasoning, or just definitions? 5) Which questions on the DQB reveal curiosity about dormancy, so that Lesson 3 can start from the learners' own words? 6) If electricity or the time-lapse videos were unavailable, did the teacher-prepared alternatives (charts and the Lesson 1 events table) work? What would you change next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed soaked Rosecoco bean seeds (testa, hilum, micropyle, two cotyledons, plumule, radicle) and maize H614 grains
----------------------------	--

	(one cotyledon called the scutellum, endosperm, coleoptile, coleorhiza, plumule, radicle). We also used Mr. Wafula's data: maize stem width 6, 14, 22 mm and shoot height 3 to 125 cm; bean stem width 2, 4, 6 mm and shoot height 4 to 44 cm.
What did I learn?	Growth is a permanent, irreversible increase in size or dry mass; development is a change in complexity and form through differentiation. Monocot (maize) and dicot (bean) seeds have different structures: the bean stores food in two cotyledons, while the maize stores food in the endosperm and one cotyledon (scutellum) absorbs it. Each part has a function, such as protection, food storage, and forming the root and shoot.
How does this explain the phenomenon?	Maize and beans differ in seed structure, which is part of why they emerge differently (maize by a coleoptile on Day 7; bean by an arched hypocotyl lifting two cotyledons on Day 8). Both seeds contain an embryo and a testa or coat, but the structure alone does not yet explain why some healthy-looking seeds did not germinate. This raises the next question: What is seed dormancy?

LESSON 3 (40 minutes (one single period; Part B scarification and soaking set-up runs in the same period, with germination checked in Lesson 4 or as a take-home observation)): Seed Dormancy: Why Do Healthy-Looking Seeds Not Germinate?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use seed specimens and a simple scarification investigation to explain why 8 healthy-looking Rosecoco bean seeds on Mr. Wafula's farm did not germinate, and how dormancy can be broken. This moves the class from the puzzle of healthy-looking seeds toward the next question: how does germination occur?
Knowledge	By the end of the lesson, learners should be able to: (1) define dormancy as a viable seed failing to germinate even in favourable conditions; (2) distinguish dormant-but-viable seeds from non-viable seeds and from seeds lost to pests or waterlogging, using the 12 non-germinating maize and 20 non-germinating bean seeds; (3) outline causes of dormancy (hard impermeable seed coat, immature embryo, chemical inhibitors such as abscisic acid); (4) outline ways of breaking dormancy (scarification, soaking, and others) and relate them to the cause; (5) explain the significance of dormancy to the plant and to the farmer.
Skills	Observing and examining seed specimens (hand lens, fingernail and water-uptake test); Planning and carrying out an investigation (NGSS SEP 3); Analysing and interpreting data (NGSS SEP 4): sorting the 12 and 20 non-germinating seeds into categories and computing the 7 of 8 result; Constructing explanations using Claim-Evidence-Reasoning (NGSS SEP 6); Engaging in argument from evidence (NGSS SEP 7); Communicating in simple, clear scientific English.
Attitudes	Learning to Learn: the learner reflects on their own work and adjusts accordingly while carrying out activities to demonstrate ways of breaking seed dormancy. Unity: the learner works harmoniously with others in the group while examining specimens and setting up treatments. Curiosity and respect for evidence when a prediction turns out to be wrong.
Key Inquiry Question	How does growth occur in plants? Lesson focus: Why did some healthy-looking bean seeds not germinate, and how can the farmer make them germinate? Pose to the class: 'If the rain, soil and planting day were the same, what is different about the seed itself?'
Purpose in Storyline	Lesson 1 raised the puzzle of healthy-looking seeds that did not germinate at Mr. Wafula's farm. Lesson 2 gave learners the structure of the maize H614 and Rosecoco seeds, including the seed coat (testa) and embryo. Lesson 3 uses that structure to explain one cause: a hard, impermeable testa makes 8 viable bean seeds dormant (7 of 8 germinated after scarification). It also sorts the other failures (12 maize, 20 beans) into viable versus not viable. Remaining failures (waterlogging, lack of oxygen) lead to Lesson 4 and the new question: 'How does germination occur?'
Safety Notes	Scarify seeds with sandpaper or a nail file, never with a razor blade or knife; rub away from fingers and hold seeds on a flat tray. Use only warm (not boiling) water for soaking; the teacher alone handles hot water for demonstration. Wash hands after handling seeds, as some stored seed may be treated with pesticide: label and keep any treated seed away from learners, use only untreated seed. Do not taste any seed. Keep seeds and water away from eyes. Clear spills to prevent slipping.

B. LESSON OVERVIEW

Madam Achieng's Form 3 class at a Kitale secondary school returns to Mr. Wafula's field: 100 maize H614 and 100 Rosecoco bean seeds were planted on the same day, 3 cm deep, after 25 mm of rain. At Day 14, 88 maize and 80 beans had germinated. Learners predict why 8 healthy-looking bean seeds sat in the soil, then examine bean and maize seeds, sort the 12 and 20 non-germinating seeds into causes, and compare untreated and scarified (nicked) seeds. They use the field result that 7 of the 8 hard-coated beans germinated by Day 7 after scarification. Learners write a CER explanation, add questions to the Driving Question Board, and revise their Day 1 model by adding dormancy as one reason seeds differ. Offline-friendly: no electricity or internet is needed; teacher-prepared printed photos and a written

summary of the Khan Academy 'Seed dormancy' reading replace the online resource where there is no connectivity.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: growth development plants seed  HTML: Seed Dormancy and Germination - Advanced (Flexbook) Source: CK-12 Search ARES for similar readings	(8 minutes) Learners look at the data card: Day 14 results out of 100: maize 88 germinated, 12 did not; beans 80 germinated, 20 did not. Mr. Wafula dug up the failed seeds and found that 8 bean seeds looked healthy, firm and unspoiled. THINK (individually, 1 minute): 'Why might a healthy-looking Rosecoco bean seed refuse to germinate when the rain, soil, depth and planting day were the same?' Learners write a prediction and a reason in their notebooks. PAIR (2 minutes): compare predictions. SHARE (3 minutes): groups give predictions, e.g. 'the seed is old', 'no water got in', 'the seed is sleeping'. Teacher records predictions on the board under headings: seed problem / environment problem / not sure. They then predict: 'If we scratch the hard outside of the seed, will it germinate?' Each learner records Yes/No and why. This is a POE prediction to be tested later.	Display the data card on the board or as a printed handout (no electricity needed). Say: 'Mr. Wafula planted on the same day, in the same field, with the same rain. Eight healthy-looking bean seeds still did not germinate. What could be different about the seed itself?' Give the Think step 1 minute of silent writing. WAIT TIME of 10-15 seconds after asking for ideas before accepting any answer; count silently. Cold-call 4 learners from different groups (not only volunteers) with: 'Wanjiru, what did you and your partner predict?' Ask a follow-up probe: 'What makes you think so?' Do NOT correct predictions; write them all. Ask: 'Which of these ideas could we test today?' Point to the Day 1 model and the DQB question 'Why do healthy-looking seeds not germinate?'. Clarify the words 'viable' (alive embryo) and 'dormant' (viable seed that fails to germinate in favourable conditions) only after predictions are shared.	Think-Pair-Share inside Predict-Observe-Explain (POE). Individual thinking makes every learner commit to an idea before seeing evidence; pairing lets learners rehearse and refine wording in simple scientific English; the whole-class share surfaces a range of ideas (old seed, no water, pests) that later evidence will sort. Written predictions let learners see for themselves later if they changed their ideas.	Oral probing and misconception check. Observable indicators: (a) learner writes a prediction with a reason, not only 'yes/no'; (b) at least 3 different ideas are recorded on the board; (c) listen for the misconception 'a seed that does not germinate is dead'; note names of learners who hold it and revisit in the Explain phase. Teacher tallies on a checklist how many learners predicted that scratching the seed would help (Yes/No).
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: growth development	(14 minutes) Part A: Examine seed specimens (6 minutes). Groups of 4-5 receive a tray with about 10 untreated Rosecoco bean seeds, 10 maize H614 seeds (or equivalents), a hand lens, sandpaper, and a cup of water. Groups: (1) observe the testa, hilum and micropyle with the hand lens; test the hardness by pressing	Before the period, prepare trays and sandpaper, with beans from the local market (e.g. Kitale or Nairobi) soaked or dry as needed. Circulate and say: 'Show me where water could enter this seed. What does the testa feel like?' and 'Fatuma, what is your group seeing under the hand lens?' Remind: 'Rub gently, away from your	Predict-Observe-Explain: the Observe step provides evidence that directly tests the predictions (does scratching the testa change water uptake and germination?). Sorting data into categories (dormant, non-viable, pests, waterlogged) helps learners see that 'not germinating' has several different causes, only one of which	Observation checklist during the practical (tick when seen): handles seeds safely and scarifies gently; labels cups correctly; records observations in a table; notices that the scarified seeds take up water or show a swollen look sooner than untreated ones; sums $5+4+3 = 12$ and $8+6+4+2 = 20$ correctly; correctly identifies 8 as

plants seed  HTML: Seed Dormancy and Germination - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	with a fingernail; (2) soak 3 seeds untreated and 3 seeds scarified (rubbed on sandpaper until a small dull patch shows, taking care not to damage the embryo) in separate labelled cups of water; (3) record in a table: seed type, treatment, appearance, hardness. Part B: Sort the non-germinating seeds (5 minutes). Using the data card with the reasons for non-germination: Maize (12): 5 eaten or rotted by fungi/pests, 4 old stored seed with dead embryos (not viable), 3 planted in a waterlogged low patch (no oxygen). Beans (20): 8 hard impermeable seed coat (dormant, healthy-looking), 6 waterlogged low patch, 4 old seed with dead embryos, 2 eaten by pests. Learners complete a table with columns: cause, maize, beans, viable (alive embryo)? dormant? and check totals 12 and 20. Part C: Field result (3 minutes): After the 8 hard-coated bean seeds were scarified (nicked) and replanted, 7 germinated by Day 7. Learners write the fraction and the percentage (7 of 8 = 87.5%, about 88%) and note what the 1 remaining seed might mean. Learners who finish early read the printed Khan Academy 'Seed dormancy' summary card (teacher-prepared, offline) and list two causes of dormancy.	fingers, only until the shiny surface turns dull.' Check group roles (reader, recorder, handler, reporter) to maintain Unity. Use WAIT TIME of 10-15 seconds after each question, e.g. 'Which of these seeds could still grow if the conditions were right?' Cold-call 3 groups to report their sorting of the beans and 2 groups for maize. Probe: 'Is a non-viable seed the same as a dormant seed? How do you know?' Use the printed copy of the table to direct learners to compare maize (0 dormant) with beans (8 dormant). Without electricity, use drawings on the board in place of video. If digital resources work, show the Khan Academy seed dormancy page or the bean time-lapse. Clarify difficult terms (impermeable, viable, scarification) on the board in simple English.	is dormancy. Comparing maize (0 dormant) and beans (8 dormant) links back to the seed structure learned in Lesson 2 (testa and embryo).	the dormant group. Peer check: partners swap tables and check totals. Oral probe: 'Which of the 20 non-germinating beans still had a living embryo?' (Expected: the 8 dormant and the 6 waterlogged seeds could be viable; the 4 old seeds were dead.) Flag learners who place the dormant seeds under 'dead'.
Explain Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: growth	(10 minutes) Learners build a Claim-Evidence-Reasoning (CER) explanation in their groups for the question: 'Why did 8 healthy-looking Rosecoco bean seeds not germinate at Mr. Wafula's farm, and why did 7 of them germinate after scarification?' Sentence frames on the board: Claim: 'The 8	Model one CER sentence set, then let groups write. Say: 'Your claim must answer the question. Your evidence must be a number or an observation. Your reasoning must say why.' Circulate and check that each group cites the numbers 8, 7 of 8 and 0 dormant maize. WAIT TIME of 10-15 seconds before	Claim-Evidence-Reasoning (CER) turns the observations and field numbers into a causal explanation, forcing learners to link data (8 dormant, 7 of 8 germinating after scarification, 0 dormant maize) to the mechanism (impermeable testa blocking water and oxygen entry). Peer challenge and Think-	Score each group CER on a 3-point rubric: Claim (answers the question), Evidence (cites at least 2 numbers from the data), Reasoning (links the hard testa to blocked water or oxygen uptake and to the effect of scarification). Observable indicators: uses the word 'viable' correctly; separates

development plants seed  EXERCISE: Embryo Growth and Development Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	seeds did not germinate because ____.' Evidence: 'The data show ____ (for example 8 beans had a hard impermeable seed coat; 7 of 8 germinated by Day 7 after scarification; maize had 0 dormant seeds).' Reasoning: 'This means ____ because the testa ____ so water and oxygen ____ and the embryo ____.' Then learners connect to ways of breaking dormancy: scarification (nicking or rubbing the testa), soaking in water for some hours, plus others (exposure to cold or light, passing through animals' digestive tracts, fire or smoke, hormone treatment such as gibberellin). They match each cause to a way of breaking it (hard testa to scarification or soaking; chemical inhibitor such as abscisic acid to leaching with water, cold or gibberellin; immature embryo to after-ripening/storage). Significance of dormancy: seeds survive dry or cold seasons and germinate when the rains come, spreading germination over time; the farmer uses scarification to get a uniform stand. Learners revise their Part 1 prediction: 'Was I right? What did the evidence change?' Then Think-Pair-Share: 'Does dormancy explain ALL the failures in the field? Which failures still need explaining?' (Expected: waterlogging, pests and dead seed; waterlogging and oxygen lead to Lesson 4.)	calling on anyone, and cold-call 4 groups to read claims, then 3 learners to challenge: 'Kiprop, do you agree with their reasoning? What evidence supports or weakens it?' Probe: 'If the testa is hard, how could the seed ever germinate in a natural field?' (Expected: weathering, microbes, animals, soil abrasion, fire.) Ask: 'Why might dormancy be useful for a wild plant but a problem for Mr. Wafula?' Correct the misconception that dormant seeds are dead or diseased. Confirm vocabulary: dormancy, viable, scarification. Do not teach germination conditions (water, oxygen, temperature) or the types of germination; say: 'That belongs to our next lesson.' Clarify the hormone idea only briefly: 'Abscisic acid helps keep seeds dormant; we meet hormones fully in Lesson 7.'	Pair-Share strengthen argument from evidence. Matching cause to remedy shows that the way of breaking dormancy depends on why the seed is dormant.	dormancy from death, pests and waterlogging; matches at least 3 of 4 causes to ways of breaking dormancy. Quick quiz (2 minutes, oral or on slips): 'Name two ways to break dormancy.' 'Is a dormant seed alive?' Peer review: one group exchanges its CER with another and gives one star and one wish.
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in</i>	(4 minutes) Learners return to the Driving Question Board started in Lesson 1 (chart paper or a wall space with sticky notes). Wanjiru's earlier question 'Why do healthy-looking seeds not germinate?' is moved into the 'Partly answered'	Say: 'Which question from our board did we answer today, and which new question did the evidence raise?' WAIT TIME of 10-15 seconds before taking suggestions. Cold-call 3 learners to read a new question and where	Discussion and questioning with the Driving Question Board: placing evidence next to questions helps learners track how each lesson contributes to the driving question, and turns gaps in understanding (e.g. waterlogging,	Exit ticket check: each learner states at least one piece of evidence with a number (e.g. 8 dormant beans or 7 of 8 germinated) and links it to differences between seeds, not to rain or planting date. Observable

the ARES library for this activity.  Search ARES for videos Search terms: growth development plants seed  HTML: Seed Dormancy and Germination - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	column with a note: 'Dormancy (hard testa) explained 8 of 20 bean failures. 7 of 8 germinated after scarification.' Learners add new questions on sticky notes, e.g. 'How does germination occur?', 'Why did the 6 beans in the waterlogged patch fail?', 'What does a seed need to germinate?', 'Does maize ever have dormancy?', 'Why did the last 1 of 8 seeds not germinate?' Each group posts at least one question and says which lesson might answer it. Individually, learners complete an exit ticket: (1) 'What evidence did we gather today?' (2) 'How does this help answer our driving question: What causes the differences in germination, emergence and growth of maize and bean plants?'	to place it. Highlight the question that leads to the next lesson: 'How does germination occur?' Sort questions: answered, partly answered, new. Say: 'Wanjiru started this question in Lesson 1, so what can we now tell her?' Collect exit tickets at the door. Read a sample at the end of the day to see who still thinks dormant seeds are dead.	oxygen, germination process) into the purpose for the next lesson.	indicators: at least one new, testable question per group; questions posted in the right column; learners can say whether a question is answered, partly answered or new. Teacher notes the number of exit tickets that correctly separate dormant from non-viable seeds.
Model Building Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: growth development plants seed  HTML: Seed Dormancy and Germination - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	(4 minutes) Learners take out their Day 1 drawing and explanation of why Mr. Wafula's maize and beans behaved differently. Using a different coloured pen, each learner revises the model by: (1) adding a labelled seed drawing of a bean with an intact, hard testa that blocks water entry (label testa, embryo, hilum, micropyle); (2) adding a box 'Dormant but viable: 8 beans, 0 maize' with an arrow 'scarification, 7 of 8 germinated by Day 7'; (3) adding a box 'Not germinated for other reasons: dead embryo, pests, waterlogging'; (4) writing one sentence: 'Seeds differ because of seed condition (dormancy, viability) as well as the environment.' They mark with a question mark what the model still cannot explain (how the seed actually starts to germinate, and why waterlogging stops germination). A few learners pin	Say: 'Open your Day 1 model. Use a new colour so we can see how your thinking has changed.' Circulate and prompt: 'Where in your model does the testa appear? What happens to the 8 seeds in your drawing?' Insist on correct labels (testa, embryo, cotyledon) from Lesson 2. WAIT TIME of 10-15 seconds before asking, 'What is still missing from your model?' Cold-call 3 learners to share one revision and one remaining question. Remind learners not to add the conditions of germination yet, as we will do that after the next lesson. Set the home task and safety reminder (use only untreated seeds; wash hands). Ask learners to bring a clean plastic cup or jar for Lesson 4.	Model revision: learners update the Day 1 drawing with new evidence, which makes visible the change in their thinking and ties dormancy to the anchoring phenomenon. Marking the unanswered parts with question marks keeps the model as a tool for the next lesson's inquiry. A short Gallery Walk (in the next lesson or at the end of this one if time permits) provides peer review.	Model check using three criteria: (1) includes dormancy with a correct cause (hard impermeable testa) and the numbers 8 and 7 of 8; (2) separates dormant from non-viable seeds and from pests or waterlogging; (3) identifies at least one unanswered question. Peer review: partners tick the criteria and give one suggestion. Observable indicator: revisions in the second colour show a real change, not just a copy of the old drawing.

	their revised drawings for a short Gallery Walk next lesson. Home task: soak 5 untreated and 5 scarified bean seeds (or use leftover seeds) in cups at home and record the water uptake after 24 hours in the notebook.			
--	---	--	--	--

D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners separate dormant seeds from dead seeds, or did many still say 'a seed that does not germinate is dead'? Which evidence changed their minds (the 7 of 8 result, the water-uptake test)? (2) Were the groups able to scarify safely and with limited sandpaper and trays, and how could sharing materials in a large class improve? (3) Did the CER explanations include numbers (8, 7 of 8, 0 maize) and the link to a blocked testa, or only general statements? (4) Did the offline printed cards work well for learners without access to the Khan Academy page or videos? (5) Which new DQB questions came up (germination, oxygen, waterlogging), and do they match Lesson 4? (6) Which learners need support with terms such as viable, impermeable and scarification? Plan to revisit these with a short word wall, and check the home soaking results at the start of the next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What we observed: Bean seed testa is hard; scarified seeds took up water faster than untreated ones. In Mr. Wafula's field at Day 14, 12 maize (5 pests/rot, 4 dead embryos, 3 waterlogged) and 20 beans (8 hard seed coat, 6 waterlogged, 4 dead embryos, 2 pests) did not germinate. 8 dormant beans looked healthy; 7 of the 8 germinated by Day 7 after scarification.
What did I learn?	What we learned: Dormancy is when a viable (living) seed fails to germinate even in favourable conditions. Causes include a hard impermeable testa, chemical inhibitors such as abscisic acid, and an immature embryo. Dormancy can be broken by scarification, soaking and other methods that match the cause. Dormancy helps wild seeds survive bad seasons but can delay germination on a farm. Not every failed seed is dormant: some are non-viable, eaten by pests or waterlogged.
How does this explain the phenomenon?	How this helps explain the driving question: Some healthy-looking bean seeds did not germinate because of a hard, impermeable testa (dormancy), while no maize seeds were dormant. So differences in germination depend on the seed's structure and condition as well as the environment, not on the rain or planting date. Other failures (waterlogging, lack of oxygen) still need to be explained, which raises our next question: How does germination occur?

LESSON 4 (80 minutes (double period); jars set up earlier in the week so Day 5 results are ready): Germination: Conditions Necessary and Types (Hypogeal vs Epigeal) in Maize H614 and Rosecoco Beans

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use germination jars and time-lapse evidence to explain which conditions are necessary for germination (water, oxygen, suitable temperature) and how hypogeal (maize H614) and epigeal (bean Rosecoco) germination differ, so they can explain why some of Mr. Wafula's seeds failed and why the two crops emerged on different days.
Knowledge	Learners should be able to: (d) investigate conditions necessary for seed germination, and (e) investigate types of germination in plants. They will state that a viable, non-dormant seed needs water, oxygen and a suitable temperature to germinate. They will distinguish germination (radicle emerges and the embryo resumes growth) from emergence (shoot appears above the soil). They will describe hypogeal germination (maize: scutellum stays underground, coleoptile pierces the soil, emergence Day 7) and epigeal germination (bean: hypocotyl arch lifts the cotyledons above ground, emergence Day 8).
Skills	Predicting from prior knowledge (POE); planning and carrying out a fair-test investigation with controlled variables; observing and recording data in a table; analysing and interpreting data (Day 5 jar counts); constructing evidence-based explanations (CER); using models and drawings to illustrate germination types; communicating in clear scientific English; collaborating in small groups with shared materials.
Attitudes	Core competency, Learning to Learn: the learner reflects on their own work and adjusts accordingly. Value, Unity: the learner works harmoniously with others while carrying out experiments to investigate conditions necessary for seed germination. Curiosity and respect for evidence, even when it contradicts a prediction.
Key Inquiry Question	How does growth occur in plants? (Lesson 4 focus: What must be present for a seed to start growing, and how do maize and beans emerge differently?)
Purpose in Storyline	Lesson 2 gave us the seed parts and Lesson 3 showed that a hard seed coat explains the healthy-looking bean seeds that did not germinate (dormancy). Now we test the remaining causes of failure for viable, non-dormant seeds: water, oxygen and temperature. We also explain why maize (emergence Day 7) and beans (emergence Day 8) appear differently even though both were planted on Day 0. This closes the germination part of the puzzle and raises the next question: How do plants show growth once they have emerged?
Safety Notes	Use only clean seeds from the school store. Do not use treated (pesticide-coated) seed; wash hands after handling seeds and soil. Seeds dressed with chemicals are not to be put near the mouth. Wipe up spilled water to avoid slipping. Use cooled boiled water (not hot water) for the no-air jar. If a fridge is not available, use a cool clay pot with wet sand as an alternative. Sharing equipment: one jar set per group of 5-6 learners.

B. LESSON OVERVIEW

Madam Achieng's class (Form 3, Kitale) returns to Mr. Wafula's puzzle: 88 of 100 maize and 80 of 100 bean seeds germinated by Day 14. Lesson 3 explained 8 bean seeds as dormant, but other seeds failed for different reasons, including 6 beans and 3 maize in the waterlogged low patch. Learners predict what a seed needs, then examine the Day 5 results from Fatuma's germination jars (20 Rosecoco bean seeds per jar): water + air + 25 C = 18 germinated; dry = 0; no air = 1; cold (4 C) = 0. They use CER to claim which conditions are necessary and link the no-air jar to the waterlogged patch. In the second half, learners watch the time-lapse videos (or use teacher-prepared drawings/flip-cards if there is no electricity), compare maize (hypogeal, emergence Day 7) and bean (epigeal, emergence Day 8), and build labelled drawings. A Gallery Walk, DQB update and Day 1 model revision close the lesson. Everything is done in small groups with local materials: maize, beans, jars, cotton

wool or cloth, sand, cooking oil.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Organism growth and the environment <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Seed Dormancy and Germination - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	(12 min) Learners recall Mr. Wafula's field: 25 mm of rain, same day, same field, yet 12 maize and 20 bean seeds did not germinate. Individually they write in their notebooks: (1) What does a healthy, non-dormant seed need to germinate? (2) Four jars, each with 20 Rosecoco beans, were set up: A) water + air + 25 C, B) dry, C) submerged in boiled cooled water with an oil layer (no air), D) wet but kept at 4 C. Predict how many seeds in each jar will germinate by Day 5 and give a reason. They then think-pair-share and record their predictions on the group chart. Next they predict how the maize and bean shoots will reach the surface: 'Will they emerge the same way?' and sketch their prediction.	Display the 4 jar set-up on the board or hold up the actual jars. Say: 'Mr. Wafula's seeds all got the same rain. Lesson 3 told us 8 beans were dormant. What else could stop a healthy seed from germinating?' WAIT TIME 10-15 seconds in silence before anyone speaks. Say: 'Write your own prediction first, no talking.' Give 2 minutes for individual writing, then 3 minutes for pairs. Cold-call 4 pairs (not volunteers): 'Pair 3, what number did you predict for the dry jar, and why?' Record all predictions on the board without evaluating them. Ask: 'Who predicted something different from their partner? What was the disagreement?' Circulate and note misconceptions such as 'seeds need light to germinate' or 'soil is needed'. Do NOT correct them yet; say: 'We will let the jars tell us.'	Predict-Observe-Explain (POE) and Think-Pair-Share: Writing predictions before seeing evidence commits learners to a position, so surprises in the data (for example, that light and soil are not necessary) create cognitive conflict and drive deeper explanation. Think-Pair-Share gives every learner a voice in large classes and lets the teacher hear the range of ideas without calling on only the confident few.	Observation checklist: Does the learner write a numeric prediction for each jar? Does the reason mention water, air or warmth? Oral probing: 'Why do you think the dry jar will have no germination?' Note misconceptions (light needed, soil needed, all seeds will germinate regardless) on a class tally. Indicator of readiness: at least 3 of 4 predictions have a stated reason.
Observe Phase  VIDEO: Organism growth and the environment <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  H5P:	(25 min) Part A (15 min): Groups (5-6) examine the prepared jars from Fatuma's set-up (20 Rosecoco seeds each, Day 5): Jar 1 water + air + 25 C; Jar 2 dry; Jar 3 no air (boiled cooled water with oil layer); Jar 4 cold (4 C). They count germinated seeds (radicle visible), record in a table: Jar 1 = 18, Jar 2 = 0, Jar 3 = 1, Jar 4 = 0. They identify the variable changed in each jar and the variables kept the same (seed variety, number of seeds, time). They also compare the 18 germinated with the 2 that	Before the lesson, set up the jars 5 days earlier (prepare a back-up set of photographs or a class chart of the results in case of failure). Say: 'Count carefully. A seed has germinated only when the radicle has come out. Two people count, one records.' Circulate with the observation checklist; ask: 'Which variable did your group change? How do you know the other jars are a fair test?' and 'Why must the number of seeds be 20 in each jar?' WAIT TIME 10 seconds after each question. Cold-call 3 groups	Predict-Observe-Explain (the Observe step) with controlled variables: Comparing jars that differ in only one condition lets learners isolate cause and effect, which is the core of a fair test. Linking the no-air jar to the waterlogged patch connects the lab model to the field phenomenon. The time-lapse gives dynamic evidence of the process that a static diagram cannot show.	Observation checklist indicators: counts germinated seeds correctly using the radicle criterion; records data in a table with units (seeds out of 20); names the changed variable in each jar; names at least two controlled variables; records the Day 7 and Day 8 emergence events correctly. Oral probe: 'Why did one seed germinate in the no-air jar?' (accept: a small amount of dissolved oxygen or residual air). Peer check: partners confirm each other's counts.

Tra's Red Bean Plant Source: Global Digital Library - Book Catalog Search ARES for similar readings	did not and suggest why (e.g., one dormant or non-viable seed). They discuss the waterlogged low patch where 6 beans and 3 maize failed. Part B (10 min): Learners watch the two time-lapse clips (Bean Time-Lapse 25 days soil cross section; Growing corn time lapse) or study teacher-prepared labelled drawing cards for Day 3, 7, 8, 10, 12 events. They record in a table: bean radicle Day 3, hypocotyl arch and cotyledons above ground Day 8, first unifoliate leaves Day 12; maize radicle Day 4, coleoptile pierces soil Day 7, first leaf Day 10, scutellum stays underground. If time permits, they uncover a few soaked seeds in sand trays to see where the cotyledons are.	to report their counts and compare with the key (18, 0, 1, 0). If counts differ, ask: 'What might explain the difference?' For the time-lapse: pause at Day 7-8 and ask: 'Where are the seed leaves of the bean? Where is the maize seed leaf?' If there is no electricity, show the drawing cards and say: 'This card replaces the video; read the days carefully.' Clarify terms: germination, emergence, radicle, hypocotyl, coleoptile, scutellum. Insist that learners use correct vocabulary: 'Say germination when the radicle comes out, emergence when the shoot appears above the soil.'		
Explain Phase  VIDEO: Organism growth and the environment <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Seed Dormancy and Germination - Advanced (Flexbook) Source: CK-12 Search ARES for similar readings	(25 min) Part A (12 min): CER on conditions. Groups write: Claim: 'Germination needs water, oxygen and a suitable temperature.' Evidence: Jar 1 = 18 of 20, Jar 2 (dry) = 0, Jar 3 (no air) = 1, Jar 4 (cold) = 0. Reasoning: water activates enzymes and softens the seed, oxygen is needed for aerobic respiration to release energy for embryo growth, and a suitable temperature (about 20-25 C) lets enzymes work. They then explain why the waterlogged low patch on Mr. Wafula's farm caused 3 maize and 6 bean seeds to fail: soil pores filled with water, no oxygen. They also state which of the failures in the farm data are NOT explained by these conditions (pests, non-viable embryos, dormancy). Part B (13 min): Learners draw and label hypogeal germination of maize H614 (scutellum stays underground, coleoptile pierces soil, emergence Day 7) and epigeal germination of	Say: 'Use the sentence frame: My claim is _____. The evidence is _____. This shows _____ because _____. Display the CER frame. WAIT TIME 15 seconds before taking responses. Cold-call 5 learners (one per group) to read their reasoning. Probe: 'Jar 3 had water and a good temperature but only 1 germinated. What does that tell us about oxygen?' and 'How is Jar 3 like the waterlogged patch on Mr. Wafula's farm?' Push for causal language: 'Why does the embryo need oxygen? What process needs it?' (aerobic respiration to release energy). Ask: 'Jar 1 gave 18, not 20. Does that disprove our claim?' (No; 2 seeds could be dormant or non-viable, as in Lessons 3 and the farm data). Address misconceptions such as 'light is needed' by asking: 'Were any jars exposed to light? What did the seeds in the dark soil do?' For Part B, circulate and correct drawings: 'Where is the scutellum?'	Claim-Evidence-Reasoning (CER): Learners must tie a general claim to specific numbers (18, 0, 1, 0) and a scientific reason, so the explanation is evidence-based, not opinion. Applying the claim to the waterlogged patch transfers the idea to the anchoring phenomenon. Comparing hypogeal and epigeal drawings supports modelling of structure-function differences between monocots and dicots.	Rubric for CER (0-3): claim stated correctly (water, oxygen, temperature); evidence uses at least 3 jar counts with correct numbers; reasoning links each condition to its role (enzymes, respiration). Indicator of misconception: learner says light or soil is a necessary condition. Peer review of drawings: partner checks labelling of scutellum, coleoptile, hypocotyl arch and cotyledons against the success criteria. Quick check question: 'Why is the maize seed leaf left underground?' (storage structure remains in the seed; the coleoptile carries the shoot up).

	Rosecoco (hypocotyl elongates as an arch, cotyledons lifted above soil, emergence Day 8). They write a comparison statement explaining why germination (Day 3 and Day 4 radicle) and emergence (Day 7 and Day 8) are different events.	Does it come above the ground?' Ask: 'Which part makes the bean emerge on Day 8 even though the radicle came out on Day 3?' Note that Lesson 2 already taught seed structure, so only refer to parts rather than re-teaching them.		
Driving Question Board (DQB) Creation  VIDEO: Organism growth and the environment <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  H5P: Tra's Red Bean Plant Source: Global Digital Library - Book Catalog  Search ARES for similar readings	(8 min) Learners revisit the DQB and match sticky notes to today's evidence. Wanjiru's question 'Why did healthy-looking seeds not germinate?' is updated: some were dormant (Lesson 3), others lacked oxygen in the waterlogged patch (6 beans, 3 maize), were non-viable, or were eaten by pests. A question about 'Why do maize and beans emerge differently?' is marked as partly answered (hypogeal vs epigeal). Groups write at least one new question, e.g., 'Once the seedling has emerged, how do stems get taller and thicker?' and 'How do plants show growth?', and add it to the board.	Say: 'Walk to the board. Find a question that today's jars or time-lapse helped answer. Move it to the answered/partly answered column and write the evidence on a new note.' Give WAIT TIME 10 seconds, then cold-call 3 learners: 'Wanjiru, which question did you move and what evidence did you use?' Ask: 'Which questions are still unanswered?' Point out that the Day 14 data show maize and beans differ in height later; draw attention to the unanswered question about growth after emergence to launch the next lesson: 'We now know how seeds start. How do plants show growth?' Record unanswered questions on a chart paper that can be carried over if board space is limited.	Discussion and questioning with the Driving Question Board: Matching evidence to questions shows learners how each lesson contributes to the driving question, and unanswered questions generate the purpose for the next lesson.	Exit-ticket style question on the board: 'What evidence did we gather today?' and 'How does this help answer our driving question?' Observable indicators: learner cites a specific count (e.g., 0 of 20 in the dry jar) when moving a question; at least one new question is about growth after emergence (height, thickness, branching).
Model Building Phase  VIDEO: Organism growth and the environment <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	(10 min) Learners return to their Day 1 drawings and explanations of Mr. Wafula's field. In pairs they revise the germination part of the model by adding: (1) water, oxygen and suitable temperature as necessary conditions, with the jar evidence; (2) the waterlogged patch as a cause of failure through lack of oxygen; (3) hypogeal (maize, emergence Day 7) and epigeal (bean, emergence Day 8) pathways; (4) a note that dormancy, pests and non-viable seeds are other causes. They use	Say: 'Use a different colour to show what you have changed since Day 1. Show me one thing you added and the evidence for it.' Give 5 minutes to revise, then WAIT TIME 10 seconds before the gallery. Gallery Walk instructions: 'Move clockwise every 2 minutes; leave one star (a strength) and one question.' Cold-call 3 groups afterwards: 'What feedback did you get? What will you change?' Prompt: 'Does your model explain why some seeds on the same farm did not germinate even though the	Model revision with Gallery Walk: Revising the Day 1 model with evidence makes learning visible, shows how explanations change when evidence accumulates, and peer feedback helps learners spot gaps and errors. Using different coloured pens helps learners reflect on their progress (Learning to Learn).	Model check: the revised model includes water, oxygen and temperature with at least two jar counts as evidence; links the waterlogged patch to oxygen; shows hypogeal vs epigeal with correct labels. Peer review: star and question are specific and based on evidence. Exit ticket: learner can state one piece of evidence and one link to the driving question; common misconception flagged for follow-up: confusing germination with emergence.

 HTML: Seed Dormancy and Germination - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	a different coloured pen so revisions are visible. Groups then do a short Gallery Walk, leaving one star and one question on two other models. Finally each learner completes an exit ticket.	rain was the same?' Collect exit tickets: (1) 'What evidence did we gather today?' (2) 'How does this help answer our driving question?' (3) 'What do you still wonder about growth?' Tell learners to bring a ruler and notebook next lesson to observe seedlings and look at how growth occurs. Where the class is large, groups can display models on desks and walk around in rotation to save space.		
--	---	---	--	--

D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners use jar data (18, 0, 1, 0) to support their claims, or did they rely on memory? (2) How many learners still thought light or soil was necessary for germination, and did the evidence change their view? (3) Did learners clearly separate germination (radicle, Day 3 and Day 4) from emergence (Day 7 and Day 8), and can they link the waterlogged patch to lack of oxygen? (4) Did the time-lapse or drawing cards work given power and connectivity? (5) Were all learners participating in the shared jar groups, and did the observation checklist show the quieter learners contributing? (6) Which DQB questions were moved, and is the question 'How do plants show growth?' strong enough to launch Lesson 5? Adjust the pacing and groupings for next time.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Jar results on Day 5 (20 Rosecoco bean seeds per jar): water + air + 25 C = 18 germinated; dry = 0; no air = 1; cold (4 C) = 0. Time-lapse: bean radicle Day 3, emergence (cotyledons above ground) Day 8; maize radicle Day 4, emergence (coleoptile pierces soil) Day 7, scutellum stays underground. Waterlogged patch: 3 maize and 6 bean seeds failed.
What did I learn?	A viable, non-dormant seed needs water, oxygen and a suitable temperature to germinate. Germination is the radicle emerging; emergence is the shoot appearing above the soil. Maize shows hypogeal germination (cotyledon/scutellum stays underground) and beans show epigeal germination (cotyledons lifted above ground by the hypocotyl arch).
How does this explain the phenomenon?	Seeds on Mr. Wafula's farm differed because planting on the same day with the same rain does not give every seed the same conditions: the waterlogged patch had no oxygen, so 3 maize and 6 bean seeds failed, while seed structure explains the different emergence days (maize Day 7, bean Day 8). Together with dormancy (Lesson 3), these show that seed condition and environment, not the rain or date, explain differences. This leaves the question: How do plants show growth after emergence?

LESSON 5 (40 minutes (single period; the stem-section drawing can be finished as homework)): Primary and Secondary Growth: Why Does Maize Thicken but Not Make Wood?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners distinguish primary growth (lengthening at apical meristems) from secondary growth (thickening by the vascular cambium in woody dicots) and use stem sections, drawings and models to explain why Rosecoco bean and Grevillea thicken differently from maize H614 on Mr. Wafula's farm.
Knowledge	Primary growth occurs at apical meristems of shoots and roots and lengthens the plant. Secondary growth occurs when the lateral (vascular) cambium divides, adding secondary xylem inward and secondary phloem outward, which increases stem girth and produces annual rings in woody dicots such as Grevillea robusta or Cypress. Maize is a monocot with no vascular cambium and no true secondary growth. Its stem width increased from 6 mm to 14 mm to 22 mm (Days 14, 28, 42) through primary growth. Bean stems stayed thin (2, 4, 6 mm) with the cambium only slightly active at the base.
Skills	Observing stem sections under a light microscope or photomicrographs; drawing and labelling biological diagrams; using models to illustrate stages of secondary growth; interpreting data tables (NGSS SEP: Developing and Using Models; Analyzing and Interpreting Data); communicating in simple, clear scientific English.
Attitudes	Unity: learners work harmoniously in small groups sharing microscopes, photomicrographs and textbooks. Learning to Learn: learners reflect on their own drawings and adjust them after peer review. Curiosity and respect for evidence when their first ideas about thickening are challenged.
Key Inquiry Question	How does growth occur in plants?
Purpose in Storyline	Lessons 1 to 4 explained why the seeds differ in germination and emergence (seed structure, dormancy, and conditions for germination). Once seedlings are growing, learners must explain why growth is not just getting taller: stems also thicken, and maize and beans do this differently. This lesson sets up Lesson 6 by raising the question: How can you measure growth in plants?
Safety Notes	Handle microscope slides and cover slips carefully and report broken glass to the teacher. Use blades or razors only under teacher supervision, cutting away from the body on a tile. Carry microscopes with two hands. If microscopes are few, use photomicrographs, teacher-prepared slides or hand-drawn stem sections. Wash hands after handling plant material. No electricity or internet is needed.

B. LESSON OVERVIEW

Madam Achieng's Form 3 class in Kitale revisits the phenomenon: Mr. Wafula planted maize H614 and Rosecoco beans on the same day, yet by Day 42 the maize stem was 22 mm wide and the bean stem only 6 mm. Learners predict why the stems differ in thickness, then gather evidence by observing stem sections under a microscope (or photomicrographs and the teacher's printed reading on primary and secondary growth). They compare a maize stem section, a young bean stem section and a Grevillea robusta (silky oak) or Cypress section with annual rings. They draw and label what they see and use drawings and simple models (layered paper or plasticine rings) to show stages of secondary growth. They use Claim-Evidence-Reasoning to explain why maize thickens without making wood. Learners add the new question 'How can you measure growth in plants?' to the Driving Question Board (DQB) and revise their model of how the two crops grow.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: growth development plants primary  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: growth development plants primary	Learners recall that Lesson 4 showed seedlings emerging and growing, and Lesson 2 showed the monocot/dicot difference. The teacher shows the stem width data: maize H614 6, 14, 22 mm and bean Rosecoco 2, 4, 6 mm (Days 14, 28, 42). Individually, learners write a prediction in their notebooks: 'Maize stems get much wider than bean stems. What happens inside a stem that makes it get thicker? Will a Grevillea tree trunk grow wider in the same way as the maize stem?' They sketch what they expect the inside of a maize stem, a bean stem and a Grevillea trunk to look like. Think-Pair-Share: they discuss with a partner and share.	Write the width table on the board. Say: "Same field, same rain, same planting day. Maize is 22 mm wide on Day 42 and bean only 6 mm. Think quietly and write for two minutes." Ask: "Does growth in height and growth in width happen in the same part of the plant?" WAIT TIME: 10-15 seconds before taking any answer. Pair learners for 2 minutes. Cold-call 4 pairs, including at least 2 who have not spoken yet. Record predictions on the board without judging them. Probe: "What do you picture inside a tree trunk? Why does a Grevillea trunk have rings?" Note common ideas such as 'the stem just stretches' or 'cells just get bigger'.	Think-Pair-Share combined with the prediction step of Predict-Observe-Explain (POE). Writing individually first commits each learner to an idea, so the later microscope evidence can challenge or confirm it. Pair talk lets weaker speakers rehearse in simple English before sharing.	Check notebook predictions: look for learners who sketch layers or rings versus those who write only 'it gets bigger'. Tally on the board how many predict that maize and Grevillea thicken the same way. Observation checklist: Did the learner write a prediction? Did the learner use the width data? Oral probe: "What makes you think so?" Misconceptions to note: thickening is only stretching; all plants make wood.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: growth development plants primary  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: growth development plants primary	In groups of 4-5 (sharing equipment), learners examine prepared or hand-cut sections: (1) a maize H614 stem, (2) a young Rosecoco bean stem near the base, and (3) a Grevillea robusta or Cypress stem with annual rings. Where microscopes are few, groups rotate through stations or use photomicrographs and printed pictures. They also read the teacher's short 'primary and secondary growth' reading (printed or from the textbook). They draw and label each section (epidermis, cortex, vascular bundles, xylem, phloem, vascular cambium, pith, annual rings/bark where visible). They note: maize vascular bundles are scattered with no cambium ring; bean bundles are arranged in a ring with a thin layer of cambium between xylem and phloem;	Before the activity, review safety and demonstrate focusing: "Start on low power, use two hands, and tell me if you see broken glass." Circulate with a checklist and ask each group: "Where are the vascular bundles? Which layer is the cambium?" and "Point to where new xylem would be added." Cold-call 3 groups to describe one difference between the maize and the Grevillea section, with WAIT TIME of 10 seconds. Use the primary growth link: "Which meristem lengthens the shoot tip? Which meristem thickens the trunk?" Clarify terms such as 'meristem', 'cambium' and 'girth' in simple English. For crowded classes, assign roles: focuser, drawer, labeller, reporter.	Predict-Observe-Explain (POE): the observation stage lets learners test their predictions directly against structural evidence. Comparing three sections side by side helps them notice patterns (scattered bundles versus ring of bundles versus thick wood) rather than memorising labels.	Observation checklist: Can the learner focus or use the photomicrograph correctly? Do drawings use a sharp pencil, a title and ruled label lines? Is the cambium correctly located between xylem and phloem in the bean/Grevillea drawings? Is it correctly shown as absent in maize? Oral probes: "What is different about how the bundles are arranged?" Peer review: groups swap drawings and give one improvement.

	Grevillea shows a thick ring of secondary xylem with visible annual rings. They record the pattern from the data: maize stem 6, 14, 22 mm but no cambium (primary growth); bean 2, 4, 6 mm with the cambium only slightly active at the base.			
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: growth development plants primary  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: growth development plants primary	Learners build a simple model of secondary growth using layered paper circles or plasticine rings (cambium as a thin ring that adds new layers inward as xylem and outward as phloem each year). They draw 3 stages of secondary growth in Grevillea (young stem with primary tissues, cambium becoming active, stem with several annual rings and bark) and count rings on the cypress/Grevillea section or photograph to estimate age. Using Claim-Evidence-Reasoning (CER), each group writes: Claim: why maize H614 has a wider stem than Rosecoco bean though it has no wood; Evidence: widths 6, 14, 22 mm vs 2, 4, 6 mm, scattered bundles and no cambium in maize, a cambium ring in bean/Grevillea; Reasoning: maize thickens by primary growth (cell division and enlargement behind the apical meristem), while only woody dicots have a vascular cambium that produces secondary xylem and phloem and annual rings. Groups then relate the findings back to Mr. Wafula's field.	Say: "Use your model to show me what the cambium adds each year, and which side becomes wood." Give a CER frame on the board: 'I claim... The evidence is... This shows that...'. WAIT TIME: 15 seconds after asking: "If maize has no cambium, how can the stem still reach 22 mm?" Cold-call 3 groups to share, and ask a different learner to restate each group's reasoning. Correct misconceptions: maize does not make true wood or annual rings; bean thickens only slightly because it is herbaceous with limited cambial activity. Do not teach hormones here; say: "We will find out what coordinates growth in Lesson 7."	Claim-Evidence-Reasoning (CER) plus modelling. CER forces learners to link observed structures and Mr. Wafula's width data to a mechanism. The physical model makes the invisible activity of the cambium visible and lets learners test whether it explains the annual rings.	Use a CER rubric: claim correct (maize thickens by primary growth, no cambium); evidence uses at least two data points or observations; reasoning names apical meristem or vascular cambium correctly. Quick check question: "Which growth adds rings?" (secondary growth). Peer review: groups exchange CER statements and models and give one 'star' and one 'step'. Note learners who say maize makes wood and re-teach using the maize section.
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library</i>	Learners return to the DQB with the driving question: 'What causes the differences in germination, emergence and growth of maize and bean plants?' Individually, they write on sticky notes or paper one thing this lesson explains (for example, 'Why maize stem is	Say: "Today we found out how stems get longer and wider. What does this help us answer on the DQB? What is still puzzling?" WAIT TIME: 10 seconds before calling. Cold-call 3 learners to read their notes and place them. Point to the question 'How can you	Discussion and questioning using the DQB. Linking each lesson to the main question shows learners how today's evidence on stem growth contributes to explaining the phenomenon, and learner-generated questions give direction to Lesson 6.	Check that notes are specific (name primary or secondary growth, cambium or meristem) and that questions are testable. Exit ticket: "What evidence did we gather today?" and "How does this help answer our driving question?" Look for mention of stem sections

for this activity.  Search ARES for videos Search terms: growth development plants primary  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: growth development plants primary	thicker without wood') and one new question. Wanjiru's question can be used as a model: 'How can you measure growth in plants?' Learners place notes next to earlier questions about growth in height and width and mark which questions are now partly answered. They record their evidence on the board.	measure growth in plants?' and say: "We said maize is taller and thicker, but how do we measure that accurately? That is our next lesson." Group similar notes together and mark answered questions with a tick.		or width data.
Model Building Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: growth development plants primary  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: growth development plants primary	Learners revisit their Day 1 drawings and explanations of how maize H614 and Rosecoco bean grow, and revise them. They add: (1) primary growth at shoot and root apical meristems lengthening both plants; (2) maize stem thickening from primary growth with scattered bundles and no cambium (6, 14, 22 mm); (3) bean/Grevillea with a vascular cambium ring giving secondary growth in woody dicots (bean 2, 4, 6 mm with slight cambial activity at the base); and (4) annual rings in Grevillea. They label revisions in a different colour and write one sentence on what changed and why. In a short Gallery Walk, groups display drawings and models and leave one comment on a sticky note.	Say: "Open your Day 1 model. Using a different colour, show what you now know about how the stems thicken." Circulate and ask: "Which of your changes came from the microscope evidence, and which from the width data?" WAIT TIME: 10-15 seconds. Cold-call 3 learners to explain one revision. During the Gallery Walk, set a 5-minute timer and require each learner to give one positive comment and one question. Collect exit tickets. If time is short, finish the model revision as homework.	Model revision with Gallery Walk peer review. Changing the Day 1 model with a different colour makes learning visible. Peer feedback exposes gaps (for example, forgetting that maize has no cambium) and prepares learners to add the missing measurement idea in Lesson 6.	Model check: does the revised model show the apical meristem, the cambium (in bean/Grevillea only), and annual rings? Does it explain why maize is thick but not woody? Peer review checklist: labels correct, evidence cited, explanation consistent with the width data. Exit ticket to collect: "What evidence did we gather today and how does it help answer our driving question?" Use responses to identify learners needing reteaching at the start of Lesson 6.

D. TEACHER REFLECTION

After the lesson, consider: Did learners distinguish primary growth (lengthening) from secondary growth (thickening) and correctly say that maize has no vascular

cambium? Which misconceptions remained (for example, 'maize makes wood', 'all thickening is secondary growth')? Were microscopes, photomicrographs or models enough for every group, and did roles keep everyone involved? Did the CER statements use the width data (6, 14, 22 mm vs 2, 4, 6 mm) correctly? Did learners tie today's evidence back to Mr. Wafula's field? What will you adjust in Lesson 6 so that learners can measure growth in plants?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed sections of maize, bean and Grevillea stems: maize had scattered vascular bundles and no cambium, bean had a ring of bundles with a thin cambium, and Grevillea had a thick ring of wood with annual rings. Maize stem width rose from 6 to 14 to 22 mm; bean rose from 2 to 4 to 6 mm.
What did I learn?	Primary growth at apical meristems lengthens shoots and roots. Secondary growth by the vascular cambium thickens woody dicots by adding secondary xylem and phloem and forming annual rings. Maize, a monocot, has no vascular cambium and no true secondary growth.
How does this explain the phenomenon?	Maize H614 is wider than Rosecoco bean because its stem thickens through primary growth, not because it makes wood. Only woody dicots like Grevillea thicken by secondary growth. This adds to the explanation of differences in growth between the two crops on Mr. Wafula's farm. New question: How can you measure growth in plants?

LESSON 6 (80 minutes (double period) or 2 x 40 minutes): Measuring Growth: Regions of Growth and Growth Curves of Maize H614 and Bean Rosecoco

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners measure growth in maize and bean seedlings, locate regions of growth using ink marks, and plot and interpret sigmoid growth curves, so as to explain why maize and beans grow at different rates even when planted on the same day.
Knowledge	Growth is a permanent, irreversible increase in size or dry mass and can be measured as height, stem width or root length. Primary growth occurs at the apical meristems, in a region of cell division followed by a region of elongation and then a region of differentiation. A growth curve is sigmoid, with lag, log (exponential) and stationary phases. For maize H614 the mean shoot height was 3, 12, 30, 62, 98 and 125 cm at Days 7, 14, 21, 28, 35 and 42, and for bean Rosecoco 4, 9, 18, 30, 40 and 44 cm. Maize grows steepest between Day 21 and 35, while bean grows steepest between Day 14 and 28 and then flattens.
Skills	Measuring with a ruler and thread, marking seedling roots and shoots at equal intervals, recording data in a table, calculating increases, plotting line graphs, interpreting curves, and arguing from evidence (Claim-Evidence-Reasoning). NGSS practices: analysing and interpreting data, using mathematics and computational thinking, and constructing explanations.
Attitudes	Unity: learners work harmoniously in small groups to mark, measure and record. Learning to Learn: learners reflect on their own graphs and correct them after peer review. Accuracy and honesty in recording data. Appreciation of farmers like Mr. Wafula who monitor crop growth.
Key Inquiry Question	How does growth occur in plants?
Purpose in Storyline	Lesson 5 showed that growth can be primary or secondary, and gave the stem width data. This lesson gives learners the tools to measure growth, find where it happens, and describe how it changes over time. The curves show that maize and beans differ in the speed and timing of growth. The curves also raise a new question: what coordinates and controls these growth patterns? That leads to Lesson 7: Do we have hormones in plants?
Safety Notes	Handle seedlings gently. Use water-based ink or a soft pencil to mark seedlings, not sharp pins or razor blades. Wash hands after handling soil. Keep walkways clear around trays and jars. If glass jars are used, handle them with care and report breakages to the teacher. Wash hands before leaving.

B. LESSON OVERVIEW

Madam Achieng's Form 3 class returns to Mr. Wafula's farm near Kitale. Learners already know the maize and bean seeds differ in structure, and that primary and secondary growth are different (Lesson 5). Now they ask: how do we measure growth, where in a seedling does growth happen, and how do maize H614 and bean Rosecoco grow over time? Learners first predict where on a root and shoot growth takes place and what a graph of height against time will look like. They then work in groups of 4 to 5 with maize and bean seedlings grown in tins or polythene tubes. They mark equal intervals with ink to find the regions of growth, and handle the height table for Days 7 to 42 with root length data. They plot growth curves, calculate increases, and identify lag, log and stationary phases. In the Explain Phase they use Claim-Evidence-Reasoning to describe how maize and bean growth differ. They add the new learning and the new question to the Driving Question Board and add growth curves to their model. No electricity or internet is needed; the time-lapse video is used only if available, otherwise the teacher's drawn time-lapse frames are used.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: growth development plants curve READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: growth development plants curve	Learners recall the Day 7 to 42 height trend from Mr. Wafula's field. Individually they sketch on a half page: (1) where on a bean root they think growth takes place (tip, middle or base), and (2) what graph of maize height against time will look like. Using Think-Pair-Share they compare with a partner and then with the class. Wanjiru asks: 'If the maize and beans were planted on the same day, why does maize become taller?' Learners write their prediction in their notebooks to compare later.	Draw a bean seedling and a maize seedling on the board. Say: 'Mr. Wafula planted maize H614 and Rosecoco beans on the same day. By Day 42 one is much taller. Where on this root do you think the new length is added? Draw your prediction.' Give 2 minutes of silent individual thinking, then say: 'Turn to your partner and explain your prediction in 1 minute.' WAIT TIME 10-15 seconds before taking answers. Cold-call 4 pairs, including at least 2 who have not spoken today, and record predictions as tallies on the board (tip / middle / base). Ask: 'What makes you think so?' Do not tell the correct answer. Say: 'Today we will test our predictions with evidence.'	Predict-Observe-Explain (POE) combined with Think-Pair-Share. Predicting before the investigation makes learners' ideas visible, so later evidence can confirm or correct them. Think-Pair-Share gives every learner a chance to rehearse ideas before speaking, including those with limited English.	Observe whether learners commit to a specific prediction (tip, middle, base) with a reason rather than leaving it blank. Count tally marks to identify the common misconception that roots grow along their whole length or from the base. Note learners who sketch a straight line rather than a curve for height against time. Use an observation checklist: 'Gives prediction with reason: Y/N'.
Observe Phase VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: growth development plants curve READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms:	Part A (about 25 min): Regions of growth. In groups of 4 to 5, learners use 5-day-old maize and bean seedlings grown in tins or polythene tubes. Fatuma and her group mark the root with water-based ink or a soft pencil at 2 mm intervals from the tip for 10 mm (or with thread-marked dots), and mark the shoot at equal intervals from the tip. They return the seedlings to damp soil or cotton wool and, where time allows, re-measure after 24 to 48 hours (the teacher has a prepared set marked two days earlier for comparison). They record which intervals have spread apart. Learners find that marks near the tip spread furthest, showing the region of elongation just behind	Demonstrate marking with one seedling: 'Mark gently, equal spaces, starting at the very tip. Why do we need equal spaces?' Circulate and prompt: 'Which marks have moved furthest apart? What does that tell you about where cells are getting longer?' Cold-call 3 groups to report which interval stretched most. Remind learners: 'Handle seedlings by the seed, not the stem.' For graphing, say: 'Choose a scale you can all agree on. Y-axis: 0 to 130 cm, with each big square 10 cm.' Check that every group has axes labelled with units (days, cm). Ask: 'Between which days did maize increase most? Between which days did bean increase most?' WAIT TIME 10-15 seconds before	Predict-Observe-Explain (Observe step) with structured data recording. Marking the seedling gives direct physical evidence of where growth occurs, and the graph turns a table of numbers into a pattern, so learners can see the shape of growth over time rather than only individual numbers.	Observation checklist: marks are equally spaced and begin at the tip; data are recorded in a table with units; increases are calculated correctly (maize 9, 18, 32, 36, 27 cm; bean 5, 9, 12, 10, 4 cm); axes are labelled with days and cm; two curves are plotted in different colours and are smooth rather than joined point to point. Oral probe: 'Which interval stretched most, and what does that say about where growth takes place?' Look out for learners who say growth occurs along the whole root.

growth development plants curve	the root tip, while marks further back stay close together. Part B (about 25 min): Measuring and recording. Otieno's group is given the class height table for Days 7, 14, 21, 28, 35 and 42, and the root length table for Days 14, 28 and 42. Maize height: 3, 12, 30, 62, 98, 125 cm. Bean height: 4, 9, 18, 30, 40, 44 cm. Root length: maize 10, 25, 40 cm and bean 8, 18, 26 cm on Days 14, 28 and 42. Learners calculate the increase in height between each pair of measurements. Maize increases: 9, 18, 32, 36 and 27 cm. Bean increases: 5, 9, 12, 10 and 4 cm. They plot both sets of heights on one graph (time in days on the x-axis, height in cm on the y-axis), using different colours for maize and bean. They then draw smooth curves and shade the lag, log and stationary phases. If the teacher has the two time-lapse videos (Bean Time-Lapse 25 days soil cross section; Growing corn time lapse), learners watch the shoot and root tips growing. Otherwise they use the teacher's drawn frames.	taking answers. Clarify terms: 'Lag phase means slow start. Log phase means fastest growth. Stationary phase means growth slows or stops.' For large classes, one marked-seedling set per group and one shared graph sheet per two learners. If a group's seedlings fail, give them the teacher's pre-marked seedlings.		
Explain Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: growth development plants curve  READING: No closely matching reading in the ARES	Learners use Claim-Evidence-Reasoning (CER) to answer: 'How do the growth of maize H614 and bean Rosecoco differ, and what does this tell us about the differences in the field?' In their groups they write a claim, cite at least 2 pieces of numerical evidence (for example: maize was 125 cm and bean 44 cm at Day 42; maize increased 36 cm between Day 28 and 35 while bean increased 10 cm; maize stem width 22 mm vs bean 6 mm at Day 42; maize root 40 cm vs bean 26 cm), and give reasoning that links	Give a CER frame on the board: 'Claim: ... Evidence: (numbers with units) ... Reasoning: This shows ... because ...'. Say: 'A claim without a number is only an opinion. Use data from your table.' Circulate and ask: 'Where is the lag phase on your curve? How do you know it is a lag?' Cold-call 4 groups to read their claims, then ask: 'Who has a different piece of evidence?' WAIT TIME 10-15 seconds. Probe misconceptions: 'Does a taller plant at Day 42 mean it grew faster all the time? Check the first 14 days.' (Learners should note that	Claim-Evidence-Reasoning (CER) plus Gallery Walk. CER forces learners to use data and reasoning rather than opinion. The Gallery Walk allows peer review so that learners compare different ways of interpreting the same curves and correct errors in their graphs.	Use a CER rubric (0-3): claim is clear; at least 2 pieces of correct numerical evidence with units; reasoning links evidence to regions of growth or phases of the sigmoid curve. Look for correct use of the terms lag, log and stationary, and for correct identification that maize is steepest between Day 21 and 35 and bean between Day 14 and 28. Peer review: sticky notes contain a specific star and question. Oral probe: 'What would the curve look like if we kept measuring after Day 42?' Note learners who say the

<i>library for this activity.</i>  Search ARES for readings Search terms: growth development plants curve	the evidence to the idea of regions of growth and the sigmoid phases (maize steepest log phase from Day 21 to 35; bean steepest from Day 14 to 28 then flattening). They also explain how marks spreading near the tip supports the idea that new cells are made at the apical meristem and then elongate. Learners compare their predictions from the start of the lesson with the evidence. Then they do a Gallery Walk: each group posts its graph and CER on the wall, and learners visit 2 other groups with sticky notes giving one 'star' and one 'question'. They revise their graphs after feedback (Learning to Learn). They also address the puzzle: 'Why does the bean slow down by Day 42 while maize is still taller?'	bean was taller at Day 7, 4 cm vs 3 cm, but maize was taller from Day 14 onward.) Clarify that the curve flattening for bean is the stationary phase and is not death. Manage the Gallery Walk: 6 minutes, with groups rotating every 2 minutes on your signal. Afterwards ask 3 learners to share one change they made after feedback. Explicitly link back: 'This explains part of the puzzle: the plants differ not only in how they germinate and emerge but also in how fast and for how long they grow.' Do not explain what controls growth yet. Say: 'What do you think coordinates this growth? Hold that thought.'		bean has stopped living instead of slowing growth.
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: growth development plants curve  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: growth development plants curve	Learners return to the Driving Question Board with the driving question: 'What causes the differences in germination, emergence and growth of maize and bean plants?' Individually they write on sticky notes: (1) one thing they now know about growth in maize and beans from today, and (2) one new question. They post them in the columns 'Germination', 'Emergence' and 'Growth'. Wanjiru reads out the question she wrote: 'What controls how fast the maize grows?' Learners sort questions and answered questions. Some earlier questions about growth, such as 'Why is maize taller?' and 'Where does a plant grow from?', are moved to 'Partly answered'. The class agrees on a new question for tomorrow: 'Do we have hormones in plants?'	Say: 'Look at the board. Which of our questions did today's evidence help answer? Move one note and tell us why.' Cold-call 3 learners to move notes and justify, with WAIT TIME of 10-15 seconds before each. Ask: 'What questions are we left with? We have measured when and where plants grow. What have we not yet explained?' Guide, without telling, toward the idea of a controlling mechanism: 'Something makes the cells at the tip divide and elongate, and something makes the bean flatten out. What might that be?' Write the new question in a different colour: 'Do we have hormones in plants?' Remind learners that this question will be investigated in Lesson 7. Keep sticky notes in groups so that large classes can post quickly: one representative per group.	Discussion and questioning with the Driving Question Board. Sorting questions by whether evidence has answered them helps learners see how each lesson contributes to the main question and builds ownership of the next step in the inquiry.	Check that each learner's sticky notes include one accurate statement of learning (for example, 'Growth in roots happens just behind the tip') and one relevant new question. Observe whether notes are placed in the right column and whether learners justify moving notes with evidence. Exit ticket: 'What evidence did we gather today?' and 'How does this help answer our driving question?'

Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: growth development plants curve  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: growth development plants curve	Learners revisit their Day 1 drawn model of Mr. Wafula's maize and beans. Using a different colour pen, they add: (1) a labelled seedling showing the region of cell division at the apical meristem, the region of elongation behind it and the region of differentiation; (2) a small sketch of the sigmoid growth curve with lag, log and stationary phases labelled; and (3) a note that maize grows faster and for longer than beans (125 cm vs 44 cm at Day 42), with maize steepest between Day 21 and 35 and bean steepest between Day 14 and 28. They add an arrow with a question mark and the question: 'What controls this growth?' Learners write 2 to 3 sentences explaining what they changed and why, linking it to evidence. They exchange models with a partner for a quick peer review.	Say: 'Take out your Day 1 model. Using your new colour, show what you now know about how the seedlings grow. Add evidence, not guesses.' Circulate and ask: 'Where on your model does the growth happen? What evidence from today shows that?' Cold-call 3 learners to explain one revision and say: 'What did you change, and what was your reason?' WAIT TIME 10-15 seconds. Encourage learners to leave a gap or a question mark for what they cannot yet explain: 'It is good to show what we do not yet know. Mark it with a question.' Give the exit ticket at the end: 'What evidence did we gather today? How does this help answer our driving question?' Collect the tickets. Set the bridge for Lesson 7: 'Growth is controlled. Do we have hormones in plants?'	Model revision. Updating the Day 1 model with a different colour makes the growth of understanding visible. Marking unanswered parts with a question creates the need for the next lesson on hormones.	Model check: the revised model includes labelled regions of growth, a labelled sigmoid curve and the maize vs bean comparison with at least one correct number. Peer review: the partner identifies one strength and one suggestion. Exit ticket responses are checked for evidence such as 'marks near the root tip moved apart' and for a link to the driving question. Quick quiz question: 'Name the three phases of a growth curve.' Identify learners who still believe growth occurs along the whole root.
--	---	--	---	--

D. TEACHER REFLECTION

Which learners could explain where growth happens using the ink-mark evidence, and which still believed the whole root grows? Were learners able to plot a smooth curve and name the lag, log and stationary phases without help, or did they simply join points with straight lines? Did the CER statements contain correct numbers with units (for example maize 125 cm vs bean 44 cm at Day 42), and did the reasoning connect to the idea of regions of growth? How well did groups share equipment and seedlings, and did large groups manage to include every learner? If there was no electricity or internet, did the teacher-drawn frames work as a substitute for the time-lapse videos? What new question did learners raise on the DQB, and how many of them linked it to a controlling mechanism before the teacher suggested hormones? What would I change to make the practical activity manageable in a 40-minute period?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Marks placed on maize and bean root tips spread apart most just behind the tip, showing growth occurs in a limited region. The height data for Days 7 to 42 showed maize H614 rising from 3 to 125 cm while bean Rosecoco rose from 4 to 44 cm. Maize increased most between Day 28 and 35 (36 cm) and bean between Day 21 and 28 (12 cm) in the table. Root length at Day 42 was 40 cm for maize and 26 cm for bean.
What did I learn?	Growth can be measured as height, stem width or root length. Growth in roots and shoots occurs at the apical meristem (cell division), followed by a region of elongation and a region of differentiation. A growth curve is sigmoid, with lag, log and stationary phases. Maize grows faster and for longer than beans, with its steepest growth between Day 21 and 35, while bean is steepest

	between Day 14 and 28 and then flattens.
How does this explain the phenomenon?	Maize and bean plants differ in growth, not only in germination and emergence, because they grow at different rates and for different lengths of time, even though they were planted on the same day in the same field. The curves show when growth is fastest and when it slows, but they do not explain what controls it. This raises the question for Lesson 7: Do we have hormones in plants?

LESSON 7 (80 minutes (double period)): Plant Hormones and Apical Dominance: What Coordinates Growth in Maize and Beans?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use a video, a reading and the apical dominance data from Mr. Wafula's beans to explain how plant hormones coordinate germination, dormancy, elongation and branching, and revise their model to explain the differences in germination, emergence and growth of maize H614 and bean Rosecoco.
Knowledge	By the end of the lesson, the learner should be able to demonstrate the role of hormones in plant growth and development: (1) auxin (IAA) is made in the shoot tip and promotes elongation and inhibits lateral buds (apical dominance); (2) gibberellins promote seed germination and stem elongation; (3) cytokinins promote cell division and lateral bud growth; (4) abscisic acid maintains dormancy; (5) ethylene promotes fruit ripening. Learners also link the apical dominance data (bean with tip intact: 1 main stem, 0 branches at Day 25; tip removed at Day 20: 3 lateral branches by Day 32) to hormone action.
Skills	Developing and using models (revising the Day 1 model); analysing and interpreting data (apical dominance results); constructing explanations using Claim-Evidence-Reasoning; obtaining and communicating information from a video and reading; asking questions for the DQB.
Attitudes	Unity: the learner works harmoniously with others in pairs and groups. Learning to Learn: the learner reflects on their own model and adjusts it accordingly. Appreciation of the importance of growth and development in plants, and curiosity about how farmers in Kitale can use hormone knowledge.
Key Inquiry Question	How does growth occur in plants? Lesson focus: What coordinates growth and development, and how can hormones explain why seeds that look healthy stay dormant and why beans branch only when the tip is removed?
Purpose in Storyline	Lessons 3, 5 and 6 showed dormancy (8 healthy-looking bean seeds with a hard coat), primary versus secondary growth, and sigmoid growth curves. Each left a question: what signals or coordinates these changes? Lesson 7 introduces plant hormones as the coordinating mechanism and gives the first major revision of the model. Lesson 8 will apply auxin ideas to rooting cuttings.
Safety Notes	Use a clean blade or scissors only under teacher supervision when demonstrating tip removal on a bean seedling. Wash hands after handling soil and plants. Do not taste any plant extracts. If the video cannot be shown (no electricity or internet), use the teacher-prepared printed storyboard and the reading summary instead. Keep walkways clear during the Gallery Walk.

B. LESSON OVERVIEW

Madam Achieng opens by asking learners to predict why Mr. Wafula's Rosecoco beans, with the shoot tip intact, had 0 branches at Day 25, while those with the tip removed at Day 20 had 3 lateral branches by Day 32. Learners watch the Khan Academy video on major plant hormones (or the teacher-prepared storyboard if offline) and read about hormone roles and functions. They complete a hormone table and use Claim-Evidence-Reasoning to explain the apical dominance data, then link gibberellins and abscisic acid to the 8 dormant bean seeds, 7 of which germinated after scarification by Day 7. They add new questions to the Driving Question Board, then revise their Day 1 model by adding hormones as a coordinating system. Maize H614 and bean Rosecoco stay at the centre throughout.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: growth development plants auxin  HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12 Search ARES for similar readings	Learners recall the maize and bean puzzle from Mr. Wafula's farm near Kitale. The teacher shows the apical dominance data on the board: bean with shoot tip intact = 1 main stem, 0 branches at Day 25; shoot tip removed at Day 20 = 3 lateral branches by Day 32. Using Think-Pair-Share, learners individually write a prediction in their notebooks: 'Why do you think removing only the tip made the bean branch?' They also predict: 'What might keep 8 healthy-looking bean seeds dormant, and what might wake them up?' Learners share with a partner and then with the class. A few draw an arrow on a bean seedling sketch showing where they think a 'signal' comes from.	Draw the data on the board before learners arrive. Say: 'Mr. Wafula's beans were the same variety, in the same field, with the same rain. Only one thing differed: the tip. Take 2 minutes to write what you think is happening.' WAIT TIME: 10-15 seconds of silence after posing the question and again after each pair finishes. Circulate and listen. Cold-call 4 pairs (not volunteers): 'Pair 3, what did you predict about the missing tip?' Record predictions on a 'Our Predictions' chart without judging. Ask: 'Is anyone's prediction different from this one?' Say: 'Today we will test our predictions with evidence from a video and a reading.'	Think-Pair-Share inside the Predict-Observe-Explain (POE) cycle. Individual thinking makes each learner commit to an idea before evidence arrives, pair talk lets learners rehearse and refine language in simple scientific English, and sharing with the class surfaces the range of ideas to revisit after the evidence. Predictions are kept so learners can compare them with their conclusions.	Observation checklist: (1) each learner writes a prediction in the notebook before talking (check 100% of the class); (2) predictions mention a cause (e.g. 'something from the tip', 'the tip blocks the side buds'); (3) learners use terms such as 'tip', 'lateral bud', 'signal'. Oral probing: 'What makes you think that?' Note misconceptions on a clipboard, such as 'the plant branches because it gets more sunlight when the tip is cut' or 'hormones are only found in animals'.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: growth development plants auxin  HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12 Search ARES for similar readings	Learners work in groups of 4-5, sharing textbooks. They (1) watch the video 'Major plant hormones (& how to remember) Control & Coordination class 10 Biology Khan Academy', or if there is no electricity or internet, listen to the teacher present the printed storyboard of the same content; and (2) read the passage on the roles and functions of plant hormones, including apical dominance. While watching and reading, each learner completes a table with the columns: Hormone Where made / acts Effect on the plant Link to maize or beans. Rows: auxin (IAA), gibberellins, cytokinins, abscisic acid, ethylene. Wanjiru writes the group's new questions on sticky notes or scrap paper for the DQB. Groups then look at the apical dominance data and the dormancy data: 8 dormant	Before the video, say: 'As you watch, listen for ONE hormone that makes stems longer, ONE that keeps seeds dormant, and ONE that helps side buds grow.' Play the video in two or three short segments of about 3 minutes each, pausing after each: 'Turn to your neighbour. What did you just learn?' If offline, use the storyboard: five large pictures, one per hormone, drawn on manila paper or chalkboard. WAIT TIME: 10-15 seconds after each pause before taking answers. Give the reading with the instruction: 'Underline the sentence that explains why a bean plant does not branch when its tip is on.' Circulate and ask: 'Where is auxin made? What does it do to the side buds?' Cold-call 3 groups to share one row of their table. Clarify difficult terms: 'inhibit' = stop or	Predict-Observe-Explain (Observe step) with guided note-making. A focused listening task directs attention to the evidence that bears on the driving question instead of passively viewing. The table connects each hormone to a function and to maize or beans, so facts about hormones are tied to the anchoring phenomenon rather than memorised in isolation.	Observation checklist: each learner's table has at least 4 of 5 hormones filled in correctly; learners can state that auxin is made in the shoot tip; learners can state that abscisic acid maintains dormancy and gibberellins promote germination; every group records at least one question for the DQB. Peer check: partners swap tables and tick correct rows. Oral probe: 'Which hormone is the odd one out because it works as a gas?' (ethylene). Look for the misconception that one hormone does one job only.

	beans, 7 germinated after scarification by Day 7.	slow down; 'lateral' = on the side; 'coordinate' = make parts work together. Ask learners to link back: 'Which hormone could be involved in the 8 hard-coated beans that stayed dormant?'		
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: growth development plants auxin  HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12 Search ARES for similar readings	Individually, then in groups, learners write a Claim-Evidence-Reasoning (CER) explanation to answer: 'Why did the Rosecoco bean with its tip intact have 0 branches at Day 25, but have 3 lateral branches by Day 32 when the tip was removed at Day 20?' They then write a second, shorter CER about dormancy: 'Why did 8 healthy-looking bean seeds not germinate, yet 7 of the 8 germinated by Day 7 after scarification?' using the ideas of abscisic acid, gibberellins, and the hard seed coat. Groups also answer: 'Why can the same hormones act in both maize H614 and beans, yet the plants grow differently?' Learners share with a Think-Pair-Share and compare to the claims from the Predict Phase.	Give a CER frame on the board: 'Claim: ... Evidence: (use the numbers) ... Reasoning: (use hormone words such as auxin, inhibits, lateral buds, cytokinin) ...' Model the sentence starter: 'The data show that ... because auxin from the tip ...' Say: 'Your reasoning must name a hormone and say what it does.' WAIT TIME: 10-15 seconds after each question before calling on anyone. Cold-call 5 learners in total across the CER discussion, for example: 'Otieno, what evidence supports your claim?' 'Fatuma, which number tells us the tip was the cause?' Press on weak reasoning: 'You said the tip stops branching. How does it do that?' Make clear that the 8 dormant seeds' problem was the hard seed coat, but that hormone balance is involved: abscisic acid maintains dormancy, while gibberellins promote germination. Do not re-teach causes and methods of breaking dormancy, which were covered in Lesson 3, and do not re-teach growth curves, which were covered in Lesson 6. Refer back to them only as the evidence. Explain: 'Seeds in the field are not the only thing influenced; growth and branching are coordinated by hormones.' Close by asking for a class consensus claim and write it up.	Claim-Evidence-Reasoning (CER). Learners must anchor each claim in the actual numbers from Mr. Wafula's farm (0 branches vs 3 branches; 7 of 8 seeds) and then supply a mechanism (auxin inhibits lateral buds, cytokinin promotes them, abscisic acid maintains dormancy). This separates what we observed from why it happens, and joins Lessons 3, 5 and 6 under one coordinating idea.	Observable indicators: the claim states clearly that the tip causes the lack of branching; the evidence quotes 0 branches at Day 25 and 3 branches by Day 32; the reasoning names auxin and its inhibiting effect on lateral buds, and cytokinin as promoting lateral buds. For the dormancy CER: the learner mentions viable but dormant seeds and abscisic acid or gibberellins. Peer review: partners use a 3-tick checklist (claim, evidence with numbers, reasoning with hormone). Exit ticket question: 'What evidence did we gather today, and how does it help answer our driving question?' Quick quiz (3 questions): name the hormone made in the shoot tip; name the hormone that maintains dormancy; state what happens to branching when the tip is removed.
Driving Question	Learners return to the Driving Question Board. Each group	Say: 'Look at our Driving Question: What causes the differences in	Discussion and questioning using the Driving Question Board.	Observable indicators: each group contributes at least one new

Board (DQB) Creation  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: growth development plants auxin  HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	places new questions on sticky notes or paper strips, such as 'How can a farmer use hormones?', 'Do roots also make hormones?', 'Can we make cuttings root faster?' They sort their questions into three groups: Answered today, Partly answered, and New. Wanjiru reads aloud a question from Lesson 2 or Lesson 3 that can now be answered, such as the question about what wakes up dormant seeds, and the class moves it to the 'Answered' column with a short evidence statement. A new question is added about using natural hormone sources on cuttings, leading to Lesson 8.	germination, emergence and growth of maize and bean plants? Which questions on the board did the hormones help us answer today?' WAIT TIME: 10-15 seconds before taking responses. Cold-call 3 learners to move a question and justify it: 'Kiprop, which question are you moving and what evidence supports it?' Praise specific connections: 'That link to apical dominance uses today's data well.' Point to the gaps: 'We know auxin matters for roots too. Hold that question for our cuttings project next lesson.' Do not answer new questions yourself; encourage learners to state how they might find answers.	Sorting questions into answered, partly answered and new shows learners that they are building understanding over time, keeps each lesson linked to the main question, and gives purpose to the next investigation (rooting cuttings).	question; at least one question is correctly moved to 'Answered' with a supporting fact from today's evidence; questions use the words hormone, auxin, branch or dormancy accurately. Teacher notes quality of questions (a factual question vs a testable 'what if' question). Oral probe: 'What would we need to do to find out?'
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: growth development plants auxin  HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Learners take out their Day 1 drawings and explanations of what causes the differences between maize and beans. Working individually first, then comparing with a partner, they revise their model using a different coloured pencil or pen. The revised model must add: (1) a hormone 'control system' with labelled arrows: auxin from the shoot tip inhibiting lateral buds (apical dominance); cytokinin promoting lateral buds; gibberellins promoting germination and stem elongation; abscisic acid maintaining dormancy; (2) the apical dominance result (0 vs 3 branches) as evidence beside the bean drawing; (3) a note on why maize H614 does not branch like beans in the data. They write a short caption: 'What I changed in my model and why.' Groups post their models for a short Gallery Walk where peers leave a comment using the frame: 'I like ...	Say: 'Your Day 1 model is a record of what you knew then. Today we add a layer: something that coordinates what the plant does. Use a different colour so we can see your thinking grow.' Distribute coloured pencils or pens where available; otherwise use pencil with a different style such as dotted lines. WAIT TIME: 10-15 seconds before releasing learners to revise, after asking: 'Where in your model could a hormone arrow go?' Circulate and ask: 'Why did you put the arrow here?' 'What evidence supports it?' Cold-call 3 learners to explain one revision. Organise a 6-minute Gallery Walk: groups rotate clockwise and leave one comment each. Collect the exit ticket before learners leave: 'Write one thing that changed in your model and one question you still have.' Remind learners that the model will be revised again in later lessons and that Lesson 8 will	Model revision with Gallery Walk peer review. Revising the Day 1 model in a new colour makes learning visible. It requires learners to add the coordinating role of hormones while keeping earlier evidence (seed structure, dormancy, conditions, growth), so that the model explains the differences in germination, emergence and growth of maize H614 and bean Rosecoco more fully. Peer comments push learners to justify each arrow with evidence.	Model check indicators: revised model includes at least 3 labelled hormones with correct functions; includes the apical dominance evidence (0 and 3 branches); arrows link hormone to effect (for example auxin to lateral buds, inhibits); caption explains at least one change. Peer review: each model receives at least 2 comments. Observation checklist: participation in the Gallery Walk, use of evidence in comments. Exit ticket: 'What evidence did we gather today?' and 'How does this help answer our driving question?' Look for answers that use numbers (0 vs 3 branches; 7 of 8 seeds).

	I wonder ... Evidence from today that could be added ...'	test auxin ideas on cuttings.		
--	---	-------------------------------	--	--

D. TEACHER REFLECTION

After the lesson, ask yourself: Did learners use the apical dominance numbers (0 vs 3 branches) as evidence, or did they only recall definitions? Which hormones were confused most often (for example auxin and cytokinin, or gibberellins and abscisic acid)? Did learners see hormones as a coordinating system rather than isolated facts? Were pairs and groups able to participate despite sharing textbooks and, where needed, the printed storyboard replacing the video? Which learners need support with scientific English terms such as inhibit, lateral and coordinate? Plan to revisit misconceptions at the start of Lesson 8, and note which learners might lead the cuttings investigation (for example Kiprop).

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We watched the video on major plant hormones and read about their roles. We looked at Mr. Wafula's bean data: with the shoot tip intact, 1 main stem and 0 branches at Day 25; with the tip removed at Day 20, 3 lateral branches by Day 32. We also recalled that 7 of the 8 dormant bean seeds germinated by Day 7 after scarification.
What did I learn?	Plant hormones coordinate growth and development. Auxin (IAA) is made in the shoot tip and promotes elongation while inhibiting lateral buds (apical dominance). Cytokinins promote cell division and lateral bud growth. Gibberellins promote germination and stem elongation. Absciscic acid maintains dormancy. Ethylene promotes fruit ripening.
How does this explain the phenomenon?	The bean with its tip intact did not branch because auxin from the apical meristem inhibited the lateral buds. Removing the tip lowered auxin, so cytokinin could promote lateral buds and 3 branches grew by Day 32. Hormones such as abscisic acid and gibberellins help explain dormancy and germination, so the differences between maize H614 and bean Rosecoco depend on seed structure, seed condition, environment and hormone coordination of growth.

LESSON 8 (80 minutes (double period) plus 5 minutes of daily observation over 21 days): Applying Hormones: Rooting Cuttings for the School Tree Nursery

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply what Lesson 7 taught about auxin and rooting to the nursery project. They run a fair test comparing natural rooting hormone sources (bulb onion, Aloe vera, cinnamon) against plain water on hibiscus or Duranta cuttings, then use the results to decide the best treatment for the school tree nursery. This extends the driving question from seeds to growing plants: hormones help explain why plants grow and develop differently, not only the rain or planting date.
Knowledge	By the end of the lesson, the learner should be able to: (1) explain that a stem cutting forms roots when auxin (IAA) accumulates at the cut base; (2) identify the independent, dependent and controlled variables in a cuttings fair test; (3) calculate and compare percentage rooting from the results (plain water 9 of 30 = 30%; bulb onion 21 of 30 = 70%; Aloe vera 24 of 30 = 80%; cinnamon 18 of 30 = 60%); (4) use CER to decide which natural source is best; (5) outline the steps for setting up a tree nursery.
Skills	Planning and carrying out an investigation (NGSS SEP 3); controlling variables; analysing and interpreting data (SEP 4); calculating percentages (SEP 5, using mathematics); constructing explanations using Claim-Evidence-Reasoning (SEP 6); developing and revising models (SEP 2); communicating scientific ideas in simple, clear English (SEP 8).
Attitudes	Core competencies: Citizenship (solving community problems by sharing seedlings from the nursery) and Learning to Learn (reflecting on own work and adjusting). Values: Unity (working harmoniously in groups). Patriotism (devotion to the community through conservation, inspired by Dr. Wangari Maathai). PCIs: Financial Literacy (planning to earn income from seedlings) and Prevention of Global Warming (trees as carbon sinks).
Key Inquiry Question	How does growth occur in plants? In this lesson: How can we use what we know about plant hormones to grow new plants, and which natural hormone source works best?
Purpose in Storyline	Lesson 7 showed that hormones coordinate growth, including auxin in apical dominance. Lesson 8 asks learners to use that idea in practice: if auxin controls growth, can we use natural sources to make cuttings root? The data (30%, 70%, 80%, 60%) are evidence that hormones influence growth. This prepares Lesson 9 (consolidating evidence for the final model) and Lesson 10 (sharing and selling the nursery seedlings).
Safety Notes	Wash hands after handling Aloe gel, onion extract and cinnamon powder. Do not taste any materials. Learners with allergies or skin sensitivity to onion, Aloe or cinnamon should wear gloves or serve as recorders. Use a clean, sharp blade or secateurs; only the teacher or a trained group member cuts stems, cutting away from the body. Keep cinnamon powder away from eyes and do not inhale it. Wipe up spilled water so the floor is not slippery. Dispose of used plant material in a compost pit.

B. LESSON OVERVIEW

Madam Achieng's Form 3 class in Kitale has followed Mr. Wafula's maize H614 and Rosecoco beans from seed to seedling. Now she asks: if hormones such as auxin coordinate growth, can we use them to grow NEW plants for our school nursery? Learners first predict, using Lesson 7 ideas, which of three natural sources (bulb onion, Aloe vera gel, cinnamon powder) will make hibiscus or Duranta cuttings root best (POE). They then plan a fair test with Kiprop's group leading the setup: 30 cuttings per treatment, with plain water as the control. They examine the 21-day results (9, 21, 24, 18 of 30 rooted), calculate percentages and write a CER claim. In the Explain phase they connect rooting to auxin and link back to the driving question. They add questions to the DQB and revise their cumulative model by adding hormones as a coordinating factor for how plants grow. Dr. Wangari Maathai's Green Belt Movement inspires the nursery. Practical work uses cheap local materials, small groups, and

no electricity or internet.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Phototropism Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the maize and bean puzzle: same field, same rain, same planting day, but different growth. Madam Achieng shows a hibiscus cutting (a stem piece with no roots) and four labelled jars: plain water, bulb onion extract, Aloe vera gel and cinnamon powder. Individually, learners write a prediction in their notebooks: 'Which treatment will give the most rooted cuttings after 21 days out of 30? Why?' using what they know about auxin from Lesson 7. They then Think-Pair-Share, rank the four treatments from best to worst and note the reasoning behind their ranking. Kiprop's group, who lead the project, display a prediction chart on the board.	Hold up the cutting: 'This stem piece has no roots. Can it become a whole new plant? Think quietly for 1 minute.' Give WAIT TIME of 10-15 seconds after each question before taking answers. Say: 'In Lesson 7 we learned that auxin is involved in rooting. Where in a cutting do you think auxin collects?' Cold-call 3 pairs to share rankings and record them on the chart: 'Fatuma and Otieno, what did you rank first and why?' Accept all predictions without judging. Ask: 'How is this like Mr. Wafula's seeds, where something other than rain and planting date made the difference?' Circulate and listen for misconceptions such as 'the cutting needs seeds to grow roots' or 'the substance is food for the cutting'.	Think-Pair-Share inside Predict-Observe-Explain (POE). Individual prediction makes each learner commit to an idea before seeing evidence, so later data can confirm or challenge their thinking. Pairing exposes alternative reasoning, and the class prediction chart gives a baseline for the Explain phase.	Observation checklist: Does the learner write a prediction with a reason (not just a guess)? Does the reason mention auxin, hormones or the cut base? Does the learner rank all four treatments? Oral probing: 'Why did you place plain water last?' Note misconceptions (for example, that the cutting uses the onion as food) and revisit them in the Explain phase.
Observe Phase  VIDEO: Phototropism Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Part A (about 30 min, hands-on): In groups of 5-6, learners set up a mini fair test. Each group prepares 4 labelled containers (plain water, bulb onion extract, Aloe vera gel, cinnamon powder) with cuttings of hibiscus or Duranta cut about 15 cm long, with a slanting cut below a node and the lower leaves removed. Because of class size, each group sets up a smaller number of cuttings per container and the class pools its results. Learners identify variables: independent (type of treatment), dependent (number of cuttings that root), controlled (same plant	Before the practical, say: 'Scientists need a fair test. What must stay the SAME in all four containers?' WAIT TIME 10-15 seconds, then cold-call 2 groups. Demonstrate the slanting cut at a node and handling safety. Walk around with the checklist and ask: 'Why is plain water in the investigation if it has no hormone?' and 'Why must every cutting be the same length?' Introduce data: 'Mr. Wafula could not blame the rain because rain was the same for all seeds. In our test, what is the same for all cuttings?' Check calculations: 'Fatuma, show us	Predict-Observe-Explain (Observe step) and fair-test reasoning. Handling the cuttings and the evidence table lets learners compare their prediction with real data. Converting counts to percentages gives a common scale, so differences between treatments become clear and are not just impressions.	Observation checklist indicators: group lists at least 3 controlled variables correctly; cuts are slanting and below a node; results are recorded in a table with correct labels; percentages are correct (30%, 70%, 80%, 60%); bar graph has labelled axes and an even scale. Oral probing: 'What is the control and what does it tell us?' Quick check: 'If 18 of 30 rooted, what is the percentage?' (60%).

	species, cutting length, soil and shade, watering, 21 days) and the control (plain water). Part B (about 20 min, data): Learners receive the full 21-day results of the class nursery (30 cuttings per treatment): plain water 9 rooted; bulb onion extract 21; Aloe vera gel 24; cinnamon powder 18. They record in a table, calculate percentage rooting (30%, 70%, 80%, 60%) and draw a bar graph on plain paper. Kiprop reads out the group's data and Otieno checks calculations. Teacher-prepared photographs or drawings of rooted and unrooted cuttings are used if fresh cuttings are not available.	how you got 80% for Aloe.' Clarify difficult terms (node, control, variable) in simple English. Do not tell learners the best treatment before they see the data.		
Explain Phase  VIDEO: Phototropism Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Learners use Claim-Evidence-Reasoning (CER) to answer: 'Which natural source is best for rooting cuttings in our nursery, and why?' Individually they write a claim (Aloe vera gel is best), evidence (24 of 30 = 80% rooted, compared with 70% onion, 60% cinnamon and 30% plain water) and reasoning (the treated cuttings rooted better than the control, and the cuttings were treated the same way except for the substance, so the substance made the difference; the natural sources probably supply or stimulate rooting hormones such as auxin at the cut base). They compare the results with their Predict-phase ranking and correct their ideas. They then discuss the limits of the evidence: one trial, one species and no direct measure of auxin. Finally they connect to the driving question: hormones help explain differences in growth, not just the rain or planting date. They also plan the nursery steps: prepare	Model one CER sentence on the board, then let learners write their own: 'A good claim answers the question. Evidence uses numbers. Reasoning links the evidence to the science idea.' After writing, say: 'Turn to your partner and compare. Does your evidence include numbers?' Cold-call 4 learners to read claims and ask 'Who agrees or disagrees? Show with thumbs.' WAIT TIME 10-15 seconds before each response. Probe: 'Plain water still rooted 9 of 30. What does that tell us?' (cuttings make some auxin themselves.) Ask: 'How sure are we that Aloe is better than onion? 24 versus 21 is a small difference.' Link to Lesson 7: 'Which hormone did we say is involved in rooting, and how did we see hormones working in the apical dominance experiment?' Bring in Dr. Wangari Maathai: 'The Green Belt Movement planted millions of trees. How could rooting cuttings help a nursery like ours?' Connect	Claim-Evidence-Reasoning (CER). Learners must justify a claim with numbers and a scientific idea (auxin), which moves them from 'what happened' to 'why'. Comparing the claim with the earlier prediction supports Learning to Learn and helps correct misconceptions.	Use a CER rubric: claim answers the question (yes/no); evidence cites at least two correct numbers (for example 24 of 30 and 9 of 30); reasoning mentions hormones/auxin and the fair test. Peer review: partners tick or circle one strength and one improvement. Oral probing for misconceptions: 'Is Aloe a seed?' 'Does a cutting need seeds to make roots?' Record learners who cannot link the evidence to auxin for follow-up.

	polythene tubes with soil, plant treated cuttings, water, shade, label, and later sell seedlings to the community.	to financial literacy: 'If we sell seedlings, why does choosing the best treatment matter?'		
Driving Question Board (DQB) Creation VIDEO: Phototropism Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12 Search ARES for similar readings	Learners return to the Driving Question Board (DQB) with the main question: 'What causes the differences in germination, emergence and growth of maize and bean plants?' Each group writes on sticky notes or paper strips: (a) one question answered in this lesson, for example 'Can hormones be used to make cuttings root?' (answered: yes, Aloe gave 80% rooting), and (b) one new question, such as 'Would Aloe work on maize or bean stems?' or 'Could the same treatment work on tree species like Grevillea?' Wanjiru places questions on the board and the class sorts them into 'answered', 'partly answered' and 'still to investigate'. Learners complete a short exit ticket.	Say: 'Look at the DQB. Which of our questions did today's evidence help answer?' WAIT TIME 10-15 seconds. Cold-call 3 learners to place a note and explain the link: 'Wanjiru, where does this question go and why?' Ask: 'How does using hormones on cuttings help answer why maize and beans grew differently?' (it shows hormones affect growth and development.) Distribute exit ticket: 'What evidence did we gather today?' and 'How does this help answer our driving question?' Collect to sort in the next lesson.	Discussion and questioning with the Driving Question Board. Sorting questions by status makes progress visible, links the nursery work to the main phenomenon and shows learners the remaining gaps that Lesson 9 will address.	Exit ticket success indicators: names at least one piece of evidence with a number (for example, 80% rooting with Aloe vs 30% with water); links to hormones or auxin; relates to the driving question. Observation: each learner contributes at least one note; notes are placed in the correct category. Group check: new questions are testable (for example, 'Which concentration works best?').
Model Building Phase VIDEO: Phototropism Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12 Search ARES for similar readings	Learners take out their Day 1 model (drawing and explanation of why Mr. Wafula's maize and beans differed) and the model revised in Lesson 7. In groups they add a new section: 'Hormones in growth and development' showing auxin at the cutting base promoting root formation, and the nursery application. They label evidence from today (Aloe 80%, onion 70%, cinnamon 60%, plain water 30%) beside the hormone arrow. They also add a one-sentence revision: 'Differences in growth can be caused by internal factors such as hormones, as well as seed structure, seed condition and environment.' Groups finish with a Gallery Walk: they post models,	Say: 'Open your Day 1 model. What does it NOT explain yet?' WAIT TIME 10-15 seconds. Prompt: 'Where would auxin go on your model? What evidence supports it?' During the Gallery Walk, set a timer and ask visitors: 'Does the model use today's data? Does it explain why cuttings root?' Cold-call 3 learners after the walk: 'What did you change after peer review?' Remind them to use scientific terms correctly (auxin, control, percentage rooting). If no wall space is available, groups swap papers at their desks. Close with: 'Next lesson we pull together all the evidence from Lessons 1 to 8 into the final model.'	Model revision with a Gallery Walk. Learners revisit their Day 1 model and improve it using new evidence, which shows how their understanding has changed. Peer review makes ideas public and encourages them to justify using evidence, supporting Learning to Learn.	Model check: revised models include (1) auxin or hormones, (2) at least one number from the cuttings data, (3) a clear link to how plants grow, and (4) a labelled change from the Day 1 model. Peer review: each model receives one star and one wish. Quick quiz (3 questions): 'Name the control in the cuttings test'; 'Which treatment gave the highest percentage rooting?'; 'Which hormone is involved in root formation?' Follow up with learners who omit evidence or misuse terms.

	rotate in 5-minute stations with sticky notes giving 'one star and one wish' and then use peer comments to revise. Kiprop's group presents a draft nursery plan, including labelled tubes, shade and watering, so that learners can see how the project connects to Lessons 9 and 10.			
--	---	--	--	--

D. TEACHER REFLECTION
After the lesson, reflect: (1) Did learners' predictions change after seeing the 30%, 70%, 80% and 60% results, and did they explain the change using auxin? (2) Could every group name at least three controlled variables, and which fair-test ideas need reteaching? (3) Did learners confuse the cuttings project with seed germination, and did they stay on the driving question about maize and beans? (4) Were allergies and materials managed safely and were enough local materials available for all groups? (5) Did group roles (Kiprop leading, Otieno measuring, Fatuma recording) include everyone? (6) What should be done to prepare for Lesson 9, including checking models and the DQB sort? Also plan for any learners who need support with percentage calculations or simple scientific English.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that hibiscus or Duranta cuttings treated with natural sources rooted better than cuttings in plain water: out of 30 cuttings, plain water 9 (30%), bulb onion 21 (70%), Aloe vera gel 24 (80%) and cinnamon powder 18 (60%).
What did I learn?	We learned that a fair test changes only one variable and keeps others the same, that percentages let us compare treatments, and that hormones such as auxin at the cut base help cuttings form roots, so hormones can be applied in a school tree nursery.
How does this explain the phenomenon?	Aloe vera gel is the best treatment because 80% of its cuttings rooted compared with 30% in the control, and the only difference was the treatment. This shows hormones affect plant growth and development, which helps answer our driving question about why plants grow differently: seed structure, seed condition, environment and hormones all play a part.

LESSON 9 (80 minutes (double period)): Consolidating the Evidence and Building the Final Model: Why Did Mr. Wafula's Maize and Beans Behave So Differently?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners synthesise evidence from Lessons 1-8 into one final, evidence-based explanation of why maize H614 and Rosecoco beans, planted on the same day in the same field on Mr. Wafula's farm near Kitale, differed in germination, emergence and growth. They finalise their cumulative model and review the Driving Question Board (DQB).
Knowledge	By the end of the lesson, learners should be able to: (1) distinguish between growth and development using the maize and bean data; (2) relate the structures of the monocot (maize) and dicot (bean) seeds to their functions and to hypogeal and epigeal emergence; (3) outline causes of seed failure, including dormancy (a hard seed coat in 8 of 100 beans), non-viability, pests and waterlogging; (4) state the conditions necessary for germination (water, oxygen, suitable temperature); (5) explain primary growth, the lack of true secondary growth in maize, the sigmoid growth curves and the roles of hormones including apical dominance.
Skills	Constructing explanations using Claim-Evidence-Reasoning (CER); developing and revising models; analysing and interpreting data (counts out of 100, heights in cm, stem width in mm); engaging in argument from evidence during a Gallery Walk; communicating in simple, clear scientific English.
Attitudes	Unity: work harmoniously with group members while consolidating evidence. Learning to Learn: reflect on their own Day 1 model and adjust it based on evidence. Appreciate that the differences come from seed structure, seed condition and environment, not from the rain or the planting date.
Key Inquiry Question	How does growth occur in plants? What causes the differences in germination, emergence and growth of maize and bean plants?
Purpose in Storyline	Lessons 1-8 each supplied one piece of evidence. Lesson 9 joins the pieces into a single explanation. Learners finalise the cumulative model, sort the DQB questions into answered and still open, and prepare for Lesson 10, where they present the explanation and plan seedling sharing and selling.
Safety Notes	No new practical work is required. Keep the jars, cuttings and seedlings from earlier lessons on a bench away from walkways. Learners wash their hands after handling soil, seeds or aloe gel. Treated seeds are not to be handled. Keep movement orderly during the Gallery Walk.

B. LESSON OVERVIEW

Madam Achieng opens with the original puzzle: Mr. Wafula planted maize H614 and Rosecoco beans on the same day (Day 0), 3 cm deep, after 25 mm of rain, yet 88 of 100 maize but only 80 of 100 beans germinated by Day 14, the bean emerged on Day 8 and the maize on Day 7, and some healthy-looking seeds never germinated. Learners predict how many of the puzzles they can now explain. They then observe by sorting evidence cards and data tables from Lessons 1-8 (germination breakdowns, jar results, heights, stem width, root length, hormone data). In the Explain phase they build a CER explanation for each puzzle and test it in a Gallery Walk. The class reviews the DQB, moving each question to Answered, Partly answered or Still open. Finally learners finalise their model by revising their Day 1 drawing, using a different colour for each change. Think-Pair-Share, POE, CER, Gallery Walk, model revision and DQB questioning are all used. No electricity or internet is needed; the time-lapse events are provided on a printed table or a teacher-prepared flip chart.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: growth development plants consolidation  HTML: Seed Dormancy and Germination - Advanced (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners take out their Day 1 drawing and explanation. They read the three puzzles on the board: (1) Why did 12 of 100 maize and 20 of 100 beans not germinate by Day 14, including healthy-looking seeds? (2) Why did the maize emerge on Day 7 and the bean on Day 8, with the bean lifting its seed leaves and the maize leaving its scutellum underground? (3) Why was the maize 125 cm and the bean only 44 cm by Day 42, and why did a bean branch only after its tip was removed? Learners think alone, then write a prediction on a sticky note or in their exercise book: 'Which of these can I now explain, and what is my best one-line reason for each?' They also predict: 'If I changed only the rain, would the differences disappear?' Then they pair with a neighbour and share (Think-Pair-Share).	Madam Achieng says: 'Mr. Wafula planted on the same day, in the same field, with the same 25 mm of rain. If the rain and the date were the same, what else could explain the difference? Think silently for 1 minute and write down your idea.' WAIT TIME: 10-15 seconds after each question before taking any answer. Circulate and note two or three Day 1 ideas that were misconceptions, such as 'the beans did not get enough rain'. Cold-call 4 pairs, one at a time, using 'Wanjiru, what did your partner say?' to encourage listening. Do not confirm or correct yet. Record the main predictions on the board under the heading 'Our predictions before we consolidate'.	Think-Pair-Share and Predict (the first step of Predict-Observe-Explain). Writing predictions before reviewing the evidence makes learners' current ideas visible, so the later comparison shows how much their thinking has changed. The pairing step lets learners practise explaining in simple scientific English before speaking to the class.	Oral probing: listen for learners who attribute the differences to rain or planting date, or who think growth means only getting taller. Observation checklist: tick whether each learner writes a prediction for all three puzzles. Indicator of readiness: at least 4 of the 6 groups can name one factor that is not rain or planting date (for example seed structure, dormancy or waterlogging).
Observe Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: growth development plants consolidation  HTML: Seed Dormancy and Germination - Advanced (Flexbook)	In groups of 5-6, each group gets a set of evidence cards covering lessons 1-8: Card A, germination counts at Day 14 (maize 88 of 100, beans 80 of 100) and the reasons for non-germination (maize 12: 5 eaten or rotted by fungi/pests, 4 old non-viable seed, 3 waterlogged; beans 20: 8 hard seed coat, 6 waterlogged, 4 non-viable, 2 pests); Card B, scarification of the 8 hard-coated beans (7 of 8 germinated by Day 7); Card C, jar results for 20 bean seeds at Day 5 (water + air + 25 C = 18; dry = 0; no air = 1; cold at 4 C = 0); Card D, time-lapse events (maize: radicle Day 4, emergence	Madam Achieng says: 'Your job is detective work. Place every card under the puzzle it helps to solve. Some cards help with more than one puzzle.' Distribute the card sets and printed time-lapse event tables (use the flip chart if the video cannot be shown, and play the Bean Time-Lapse 25 days soil cross section and Growing corn time lapse only where available). Circulate with probing questions: 'Which cards show that dormancy is not the same as death?' 'If the seed coat is the problem, why did 7 of 8 germinate after nicking?' 'Look at the stem width. Does the maize get thicker the same way as	Predict-Observe-Explain (the Observe step) with data sorting. Sorting cards under a puzzle requires learners to judge what each piece of evidence says, so they move from isolated lesson facts to a connected network of evidence. The checks on totals build habits of data accuracy.	Observation checklist: Did the group place Card B and Card C correctly (for example, scarification evidence under dormancy)? Did learners read the data using units (cm, mm, days)? Quick oral check: ask any learner 'Out of 100 beans, how many failed because of the hard seed coat?' (Answer: 8.) Look for the misconception that all 20 non-germinating beans were dead; the correct answer is that 8 were viable but dormant.

<i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Day 7, first leaf Day 10, 3 leaves Day 17; bean: radicle Day 3, emergence Day 8, first true leaves Day 12, trifoliate leaves Day 18, branching by Day 25); Card E, heights at Days 7-42 (maize 3, 12, 30, 62, 98, 125 cm; bean 4, 9, 18, 30, 40, 44 cm), stem width at Days 14, 28 and 42 (maize 6, 14, 22 mm; bean 2, 4, 6 mm) and root length (maize 10, 25, 40 cm; bean 8, 18, 26 cm); Card F, hormone evidence (apical dominance: intact tip = 0 branches at Day 25, tip removed at Day 20 = 3 branches by Day 32; gibberellin-soaked seeds germinate sooner; abscisic acid maintains dormancy) and Card G, the cuttings results (water 9, onion extract 21, aloe 24, cinnamon 18 rooted out of 30). Each group sorts the cards under the three puzzles, and checks the sums ($5+4+3=12$ and $8+6+4+2=20$). Fatuma's jars, Otieno's height table and Kiprop's cuttings data can be shown at the bench for learners to check against the cards. Each learner records in a table: 'What did we observe?' and 'Which puzzle does it help with?'	the bean?' WAIT TIME of 10-15 seconds after each question. Cold-call 3 groups to report one connection that they found between two cards. Correct any mistakes in the arithmetic. Make sure learners note that waterlogging affected both crops (3 maize and 6 bean seeds) because it limits oxygen.		
Explain Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: growth development plants consolidation  HTML: Seed Dormancy	Each group writes three CER explanations, one for each puzzle. Claim: a one-sentence answer. Evidence: at least two data points from the cards. Reasoning: the biological idea that links the data to the claim, using correct terms. Example for puzzle 1: Claim: 'The seeds failed for different reasons, and the healthy-looking ones were dormant, not dead.' Evidence: '8 of 20 non-germinating beans had a hard seed coat, and 7 of those 8 germinated after scarification.' Reasoning: 'A hard impermeable	Madam Achieng models one CER using puzzle 2 on the board, thinking aloud: 'My claim is that maize emerges earlier because... what evidence do I have?' Then she says: 'Now write your own. Evidence must have a number and a unit.' Provide sentence starters: 'The data show that...', 'This happens because...'. During the Gallery Walk, instruct 'Read, do not talk, for the first minute; then leave a sticky note with a question or praise.' WAIT TIME 10-15 seconds before calling on anyone.	Claim-Evidence-Reasoning (CER) with a Gallery Walk. CER requires learners to connect specific data to a biological mechanism rather than just state facts. The Gallery Walk gives each explanation a real audience, and peer questions reveal gaps in reasoning before the final model is drawn.	Peer review of the CER sheets with a short rubric: (1) a clear claim, (2) at least two data points with units, (3) correct scientific reasoning. Success indicator: at least 80% of groups include '8 of 20' or 'scarified, 7 germinated' for puzzle 1, and 'apical dominance, 0 vs 3 branches' for the branching question. Oral probing: listen for 'rain' as the explanation, which signals a lingering misconception. Teacher notes which groups confuse germination and emergence.

and Germination - Advanced (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	testa stops water entering, so the viable embryo stays dormant even in favourable conditions; the other failures were pests, non-viable embryos and waterlogging, which removes oxygen.' Learners then hold a Gallery Walk: groups post CER sheets, seed structure drawings and growth curve sketches on the walls, and rotate every 3 minutes with sticky notes marked 'Strong evidence', 'Needs a reason' or 'Question'. They then return to their own sheet and improve it. The class also discusses: 'Why do we say the differences came from seed structure, seed condition and environment, and not the rain?'	Cold-call 5 learners in total across groups with questions such as: 'Kiprop, which evidence is the strongest for your claim and why?' 'Fatuma, what is the role of oxygen in germination?' Press for reasoning: 'How do we know it was not the rain?' Clarify terms: germination (radicle emerges), emergence (shoot appears above the soil), growth (permanent irreversible increase in size) and development (changes in form or complexity). Make sure learners link the hypogeal (maize, scutellum stays underground) and epigeal (bean, cotyledons lifted) patterns to the structure of the seeds. For puzzle 3, make sure they discuss the maize growth: steepest between Day 21 and 35 (increases of 32 and 36 cm per week), slowing after Day 35 (27 cm); the bean steepest between Day 14 and 28 (9 and 12 cm per week) and flattening (10, then 4 cm); maize has no vascular cambium, so its thickening (6 to 22 mm) is from primary growth, while the bean base thickens slightly as the vascular cambium becomes active. Connect to hormones: auxin from the apical bud inhibits lateral buds, and cytokinins promote them (0 branches with the tip intact, 3 after removal); gibberellins promote germination and abscisic acid maintains dormancy.		
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library</i>	The class reviews the Driving Question Board built in Lesson 1 and updated in later lessons. Wanjiru reads each question aloud, and the class decides with a show of hands and a justification whether it is Answered, Partly answered or Still open, then	Madam Achieng says: 'Let us return to Wanjiru's first questions. Can we answer them now, and which card is our proof?' Read each question and WAIT 10-15 seconds before asking for a verdict. Cold-call 4 different learners to justify moving a card,	Discussion and questioning using the DQB. Sorting questions by status makes progress visible and shows learners that science is cumulative, and that new questions come from answered ones. It ties every earlier lesson back to the main question: What	Exit ticket with two prompts. Success criteria: the first answer names a specific data point and the second links it to the driving question. Oral probing during the DQB review: can learners give the evidence for moving a card? Quick quiz check (3 minutes, optional):

for this activity.  Search ARES for videos Search terms: growth development plants consolidation  HTML: Seed Dormancy and Germination - Advanced (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	moves the card to the matching column with a short evidence note (for example, 'What is growth and development?' is answered with the maize and bean stem width and the Lesson 2 definitions). Learners write a new question for the open column if their thinking has moved on (for example, 'How do farmers in Kitale reduce losses from waterlogged patches?' or 'Can gibberellin be used to speed up germination of old bean seed?'). Each learner completes an exit ticket: 'What evidence did we gather today?' and 'How does this help answer our driving question?'	and ask 'Who has a different view?' for at least 2 cards. Ask: 'Which question is only partly answered, and what would we need to find out?' Record unresolved questions for Lesson 10, such as the cost and income of the nursery. Collect the exit tickets and note learners whose answers list activities rather than evidence (for example, 'we did jars' instead of '18 of 20 germinated with water, air and warmth').	causes the differences in germination, emergence and growth of maize and bean plants?	'Name two conditions necessary for germination; what is dormancy; which hormone maintains dormancy?'
Model Building Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: growth development plants consolidation  HTML: Seed Dormancy and Germination - Advanced (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners take their Day 1 drawing and explanation. Using a different coloured pen, they revise it into a final model for Mr. Wafula's field. The model must include: (1) the maize and bean seed structures (scutellum and endosperm, coleoptile, coleorhiza versus two cotyledons, testa, micropyle, hilum, plumule, radicle); (2) the conditions for germination (water, oxygen, suitable temperature) with the waterlogged patch; (3) the causes of failure by crop (maize 5, 4, 3 and beans 8, 6, 4, 2); (4) hypogeal and epigeal emergence; (5) growth: primary growth at apical meristems, a sigmoid curve for each crop, stem thickening, and secondary growth in the bean base but not in maize; (6) hormones: auxin, gibberellins, cytokinins, abscisic acid and apical dominance. Each learner adds a short caption: 'My Day 1 idea was..., and now I think... because...' Groups swap models for peer review, tick the checklist	Madam Achieng says: 'Do not erase your Day 1 model. Use a new colour so that you and I can see how your thinking has grown.' Share the model checklist on the board. Circulate to check that learners show the cause of each type of failure separately and do not blur dormancy and non-viability. Ask: 'Where on your model does the rain appear, and what does your evidence say about its role?' WAIT TIME 10-15 seconds. Cold-call 3 learners to explain one change from their Day 1 model. Remind learners that the outcome is: the seeds differ because of seed structure, seed condition and environment, not because of the rain or the planting date, and that hormones coordinate growth once seedlings are established. Close by previewing Lesson 10: 'Next lesson you will present this explanation and plan how to share and sell seedlings from our school nursery for conservation and	Model revision. Comparing the final model with the Day 1 model lets learners see their own conceptual change, which supports Learning to Learn. Peer review of the models adds accountability and exposes missing links, such as the role of oxygen or the difference between maize and bean thickening.	Model check using the checklist: all six components present, correct labels, causes of non-germination separated by crop with numbers adding to 12 and 20, and a written reason for each change. Peer review sign-off. Success indicator: at least 80% of learners correctly show dormancy (hard seed coat) as different from non-viability and explain apical dominance with the 0 versus 3 branches data. Observation checklist for unity: all group members contribute.

	and suggest one improvement. Fast finishers add a note about how a farmer could improve germination: scarify hard-coated bean seed, use viable seed, avoid waterlogged patches, and protect seed from pests.	income.'		
--	--	----------	--	--

D. TEACHER REFLECTION

After the lesson, ask yourself: (1) Could most learners separate dormancy, non-viability, pests and waterlogging with the right counts? (2) Did learners use evidence with units, or did they fall back on 'because of the rain'? (3) Did any group struggle to link hormones (apical dominance, gibberellin, abscisic acid) to the field observations? (4) Did the DQB review show new, deeper questions? (5) Which learners still confuse germination with emergence, or growth with development, and what small-group support do they need before Lesson 10? (6) Were the materials (cards, flip chart, chalkboard, printed tables) enough for the class size, and how would you adapt the group size next time? Note any learners who need extra help in explaining in simple scientific English.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that on Mr. Wafula's farm near Kitale, maize H614 (88 of 100 germinated by Day 14, emerged Day 7, 125 cm by Day 42) and Rosecoco beans (80 of 100 germinated, emerged Day 8, 44 cm by Day 42) behaved differently even though they were planted on the same day after 25 mm of rain. Non-germinating seeds: maize 12 (5 pests/fungi, 4 non-viable, 3 waterlogged) and beans 20 (8 hard seed coat, 6 waterlogged, 4 non-viable, 2 pests). After scarification, 7 of the 8 dormant beans germinated.
What did I learn?	We learned that the seeds differ in structure (maize is a monocot and hypogeal; bean is a dicot and epigeal), that germination needs water, oxygen and a suitable temperature, that dormancy means a viable seed fails to germinate even in good conditions, that primary growth at meristems gives a sigmoid growth curve, that maize has no vascular cambium and no true secondary growth, and that hormones (auxin, gibberellins, cytokinins, abscisic acid) coordinate growth, including apical dominance.
How does this explain the phenomenon?	The differences came from seed structure, seed condition (dormancy, viability, pests) and the environment (such as the waterlogged patch), not from the rain or the planting date. Once growing, hormones coordinate growth: the maize grows taller and thicker through primary growth, while the bean branches only when the tip is removed, because auxin from the apical bud inhibits the lateral buds.

LESSON 10 (80 minutes (double period)): Presenting Our Final Explanation and Sharing Seedlings with Kitale

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners communicate their final Claim-Evidence-Reasoning explanation of why Mr. Wafula's maize H614 and Rosecoco beans differ in germination, emergence and growth, compare it with their Lesson 1 model, close out the Driving Question Board, and plan how to share and sell nursery seedlings for conservation and income.
Knowledge	By the end of the lesson, the learner should be able to: (1) explain that the maize and beans differ because of seed structure (maize monocot, hypogeal; bean dicot, epigeal), seed condition (8 dormant beans with hard coats, non-viable seeds, pest damage) and environment (waterlogged patch, no oxygen), not because of the rain or planting date; (2) explain that hormones (auxin, gibberellins, cytokinins, abscisic acid) coordinate growth, and that primary growth at meristems lengthens shoots and roots while apical dominance explains why beans branch only when the tip is removed; (3) calculate the sales of 60 seedlings at KES 20 each = KES 1,200, less cost of KES 300, giving a profit of KES 900.
Skills	NGSS Science and Engineering Practices: Constructing Explanations (CER), Engaging in Argument from Evidence (peer review), Developing and Using Models (comparing Lesson 1 and final models), Obtaining, Evaluating and Communicating Information (gallery walk and presentation), Using Mathematics (profit calculation). Communication in simple, clear scientific English.
Attitudes	Values: Unity (working harmoniously in groups to prepare and present); Patriotism (devotion to the community by sharing seedlings for environmental conservation). Core competencies: Citizenship (solving community problems by sharing seedlings from the nursery); Learning to Learn (reflecting on own work and adjusting accordingly). PCIs: Financial Literacy (generating income by selling seedlings); Prevention of Global Warming (establishing a tree nursery and distributing seedlings in the community).
Key Inquiry Question	How does growth occur in plants? Driving question: What causes the differences in germination, emergence and growth of maize and bean plants?
Purpose in Storyline	This is the closing lesson of the storyline. Lesson 9 produced the final model by synthesising evidence from Lessons 1-8, and Lesson 8 produced the rooted cuttings for the nursery. Here learners present the explanation to peers, show how their thinking has changed since Lesson 1, finish the DQB, and turn their knowledge into community action, following the example of Dr. Wangari Maathai and the Green Belt Movement.
Safety Notes	Wash hands after handling soil, seedlings and polythene tubes. Carry seedling trays with two people. Wear closed shoes in the nursery area. Handle cash in the mock sale only under teacher supervision and use play money or tally sheets. Do not taste any seeds, which may be treated with chemicals.

B. LESSON OVERVIEW

Madam Achieng's Form 3 class returns to the puzzle from Mr. Wafula's farm near Kitale: maize H614 and Rosecoco beans were planted on the same day, 3 cm deep, after 25 mm of rain, yet 88 maize and 80 bean seeds out of 100 germinated, beans emerged on Day 8 and maize on Day 7, and growth patterns differed. Learners first make a prediction about which parts of their Lesson 1 model were right. They then observe, in a gallery walk, the final explanations of other groups, and use a CER frame to explain the differences. Next they compare their Day 1 model with the final model and complete the DQB. Finally they plan how to share and sell the nursery seedlings: 60 seedlings of the best treatment (Aloe vera gel, 24 of 30 rooted, 80%) at KES 20 each gives KES 1,200, less KES 300 cost, a profit of KES 900. This links science to conservation, income and the example of Dr. Wangari Maathai.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: growth development plants claim READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: growth development plants claim	(10 min) Think-Pair-Share. Learners look at their Lesson 1 drawing and explanation of why Mr. Wafula's maize and beans behaved differently, which are pinned in their books or on the wall. Individually (2 min) they write a quick prediction: 'Which ideas in my Day 1 model do I think were correct, which were incomplete, and what would I now add?' They also predict: 'If our school sells 60 seedlings at KES 20 each, how much money will we make, and what will we spend it on?' They pair up with a partner (3 min), then 4 pairs share with the class. Expected ideas: 'I thought the rain or planting date caused the difference', 'I did not know about dormancy', 'I did not think about waterlogging or oxygen'.	Greet the class: 'Habari za asubuhi. Eight weeks ago we met Mr. Wafula's farm in Kitale. Same day, same field, same 25 mm of rain. Today we finish the story.' Display the phenomenon: 88 of 100 maize and 80 of 100 bean seeds germinated at Day 14. Ask: 'What did you think caused the difference on Day 1?' Give individual think time and WAIT TIME of 10-15 seconds before anyone speaks. Circulate and note 3 learners with different Day 1 ideas. Cold-call 4 pairs (not volunteers only): 'Wanjiru and partner, what was one idea you now want to change?' Do not correct predictions yet; record on the board under 'Day 1 ideas'. Say: 'Keep these ideas. At the end of the lesson we will see how far your thinking has moved.'	Think-Pair-Share: individual thinking first ensures every learner commits to a position, pair talk lets quieter learners rehearse in a low-risk setting, and class sharing exposes the range of Day 1 ideas. This creates a clear baseline against which learners measure their growth in understanding by the end of the lesson.	Observation checklist: tick learners who (a) write a specific prediction not just 'it will change', (b) name at least one earlier idea such as rain or planting date, and (c) explain their reasoning to the partner. Oral probe: 'What made you think that on Day 1?' Indicator of misconception to note: learners still attributing differences to rain or planting date, or thinking healthy-looking seeds must germinate.
Observe Phase VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: growth development plants claim READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings	(25 min) Gallery Walk. Each of 6-8 groups of 5 learners (sharing materials) has prepared a poster or manila sheet in Lesson 9 showing: (a) labelled drawings of maize and bean seed structure (maize: scutellum, endosperm, coleoptile, coleorhiza; bean: two cotyledons, testa, micropyle, hilum, plumule, radicle); (b) the germination data (88 maize vs 80 beans out of 100, and the reasons for the 12 and 20 non-germinating seeds); (c) the conditions tests (jar results 18, 0, 1, 0); (d) growth data (heights, stem widths, root lengths and sigmoid curves); (e) hormones and apical dominance (0 vs 3 branches); and (f) their final CER explanation. Groups rotate every 4	Before the walk, show the review card and the success criteria: 'A strong explanation names the claim, uses at least 3 pieces of numerical evidence, and explains the biology (reasoning).' Set a timer for 4-minute rotations and call 'Move on' clearly. As groups rotate, listen for accurate use of terms: germination (radicle emerges), emergence (shoot above soil), dormancy (viable seed that fails to germinate), viable. Ask presenters probing questions: 'Fatuma, why did 18 seeds germinate in the jar with water, air and 25 C but only 1 without air?' 'Otieno, how do your height data show maize is taller from Day 14 onward?' 'Kiprop, why did aloe	Gallery Walk with peer review: viewing multiple representations of the same phenomenon lets learners test their own explanation against others' evidence. Giving and receiving feedback with the criteria 'claim, evidence, reasoning' makes the standards for good scientific explanation visible and builds argumentation from evidence.	Peer review cards collected and checked: indicators are that comments cite specific data (e.g. 'Your bean jar numbers 18, 0, 1, 0 support the need for water, air and warmth') rather than general praise. Observation checklist for practical communication skills: speaks clearly, uses labelled diagrams, uses correct terms such as scutellum, coleoptile, testa, epigeal, hypogeal. Teacher records misconceptions such as confusing germination with emergence, or saying maize has secondary growth.

Search terms: growth development plants claim	minutes. Each learner carries a peer review card to record at each poster: one thing that is well supported by evidence, one question, and one suggestion. One presenter stays with each poster to explain it (rotating roles so that every member presents) while others visit. Learners check that numbers match the Mr. Wafula data: e.g. $5 + 4 + 3 = 12$ maize failures; $8 + 6 + 4 + 2 = 20$ bean failures; 7 of 8 scarified beans germinated by Day 7.	perform best in the cuttings?' Give WAIT TIME of 10-15 seconds after each question. Cold-call at least 6 learners across the room. Where the data are wrong, do not give the right answer: ask 'Which table in the unit can you check?' If the electricity or the videos are not available, use the teacher-prepared stills of the bean and maize time-lapse (Bean Time-Lapse 25 days soil cross section; Growing corn time lapse) printed on A4 sheets placed beside the posters. Record 3 common errors for use in the next phase.		
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: growth development plants claim  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: growth development plants claim	(20 min) Claim-Evidence-Reasoning. Back in groups, learners finalise and present a 2-minute oral CER answer to the driving question: 'What causes the differences in germination, emergence and growth of maize and bean plants?' Claim: 'The differences are caused by seed structure, seed condition and environment, not by rain or planting date.' Evidence: they cite at least three data points, e.g. (1) Day 14 germination: maize 88, beans 80 out of 100; (2) non-germination causes: maize 5 pests/fungi, 4 non-viable, 3 waterlogged; beans 8 hard seed coat (dormant), 6 waterlogged, 4 non-viable, 2 pests; 7 of 8 dormant beans germinated after scarification; (3) emergence: maize coleoptile pierces the soil Day 7 (hypogeal, scutellum stays underground) and beans raise cotyledons above ground Day 8 (epigeal); (4) heights at Day 42: maize 125 cm vs bean 44 cm; stem width 22 mm vs 6 mm; (5) apical dominance: 0 branches with tip, 3 branches after the tip is	Display the CER frame: 'I claim that ... The evidence is ... This shows that ... because ...'. Say: 'Your explanation must be one that Mr. Wafula could use to improve his next planting.' Select 4 groups to present (each 2 min), choosing different learners than in the gallery walk. After each, ask the class: 'Who can add evidence?' and 'Who can challenge the reasoning?' Give WAIT TIME of 10-15 seconds. Use these challenge questions, cold-calling 5 learners: 'If Mr. Wafula planted in the waterlogged low patch again, which seeds fail and why?' 'Why were 8 beans healthy-looking but did not germinate, and what should he do?' 'If he wanted branching in beans, what would you tell him?' 'Why does maize keep thickening without cambium?' Press for mechanism: 'You said hormones cause it; which hormone, and what does it do?' Correct remaining misconceptions explicitly using the data, for example if a learner says maize has secondary growth: 'What does the stem width data of maize, 6,	Claim-Evidence-Reasoning: separating the three components forces learners to ground each claim in the Mr. Wafula data and justify the link with biological principles. Oral defence and challenge questions move learners from description to mechanistic explanation, and shared critique refines the class consensus.	Use a CER rubric with indicators: Claim is clear and answers the driving question (0-2); Evidence includes at least 3 accurate figures from the data (0-2); Reasoning links to structure, dormancy, oxygen, meristems or hormones (0-2). Oral probing identifies misconceptions about secondary growth in maize and confusion between dormant and non-viable seeds. A quick 3-question quiz on the board (cold-call): 'Define dormancy', 'Name the hormone that maintains dormancy', 'Which type of germination is bean?' Expected answers: viable seed that fails to germinate in favourable conditions; abscisic acid; epigeal.

	removed at Day 20. Reasoning: they link the evidence to biology: the hard testa stops water uptake in dormant bean seeds; oxygen is needed for respiration so waterlogged seeds fail; the bean has two cotyledons that lift above ground while maize has a single scutellum that stays below; primary growth occurs at apical meristems, with maize's steepest log phase between Day 21 and 35 and bean's between Day 14 and 28; auxin from the tip inhibits lateral buds while cytokinins promote them; gibberellins promote germination while abscisic acid maintains dormancy. Each group then answers a 'what if' challenge from the teacher.	14, 22 mm, come from, and which meristem is missing?' Remind learners to use 'germination' and 'emergence' exactly. Allow learners to speak in simple, clear English; clarify terms such as scarification and viable.		
Driving Question Board (DQB) Creation  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: growth development plants claim  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: growth development plants claim	(10 min) Learners return to the Driving Question Board. Wanjiru and a partner read out the Day 1 questions one by one (e.g. 'Why did some healthy-looking seeds not germinate?', 'What is growth and development?', 'Why did beans branch?'). Groups move sticky notes into three columns: 'Answered with evidence', 'Partly answered', and 'New questions'. For each answered question, a learner writes the key answer and the lesson where the evidence was found (e.g. dormancy and 7 of 8 germinating after scarification in Lesson 3; conditions in Lesson 4; sigmoid curves in Lesson 6; apical dominance in Lesson 7). Learners add at least one new question for the future, such as 'Which tree species would suit Kitale best for the nursery?' or 'How do other hormones such as ethylene affect fruit ripening on a farm?'	Say: 'This board has followed us from Lesson 1. Let us see whether we have answered Wanjiru's questions.' Read each question aloud and ask, 'Is it answered? What is your evidence?' Cold-call 6 different learners. WAIT TIME 10-15 seconds. Probe: 'Which lesson gave us that evidence?' Where an answer lacks evidence, leave the note under 'Partly answered' rather than telling them the answer. Praise strong questions. Ask: 'Which question is the most important for farmers around Kitale, and why?' Take a photo or copy the final DQB onto manila paper for the class record if no camera is available. Link new questions to future learning.	Discussion and questioning with the DQB: revisiting the original questions shows learners how each lesson contributed to the main question, makes the coherence of the storyline visible, and helps learners see what science still leaves open.	Indicators: learners place questions in the correct column and justify with evidence from a particular lesson; at least 3 new, testable questions are posted. Observation: Wanjiru and others use the board to link lessons instead of recalling facts in isolation. Misconceptions flagged by notes placed incorrectly as 'Answered' without evidence.
Model	(15 min) Model revision and	Say: 'Place your Day 1 drawing	Model revision: placing the Day 1	Model check indicators: the final

Building Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: growth development plants claim  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: growth development plants claim	community action planning. (a) (7 min) Each learner places their Day 1 drawing and explanation beside their final model from Lesson 9 and writes a short comparison: 'What I drew on Day 1', 'What I now know', 'The evidence that changed my mind'. They use a table with columns: Seed structure, Seed condition (dormancy, viability), Environment (water, oxygen, temperature), Growth type, Hormones. They label added features such as scutellum, coleoptile, testa, micropyle, shoot apical meristem, auxin from the tip inhibiting lateral buds, and a note on waterlogging in the low patch. (b) (8 min) In their groups they plan how to share and sell the school nursery seedlings in Kitale. Using the nursery results from Lesson 8 (30 cuttings per treatment: plain water 9 rooted, bulb onion 21, cinnamon 18, aloe vera gel 24 rooted, 80% best), they decide to sell 60 seedlings of the aloe treatment at KES 20 each. They calculate: income $60 \times 20 = \text{KES } 1,200$; cost of polythene tubes and soil KES 300; profit $1,200 - 300 = \text{KES } 900$. They decide how to use the profit (for example, buying more tubes and soil for the next batch, or giving some seedlings free to a primary school), who to sell to (parents, farmers, church, market in Kitale), how to advertise a short pitch ('Plant a tree, protect our climate, as Wangari Maathai taught us'), and how to teach buyers to care for the seedling (watering, avoiding waterlogging, protecting from pests). They connect the plan to preventing global warming, as trees absorb carbon dioxide.	next to your final model. Put a tick where you were right and a star where you changed.' Circulate and ask: 'What evidence changed your idea about the rain?' 'Where does your model show the dormant beans?' 'What does your model explain that your Day 1 model could not?' WAIT TIME 10-15 seconds. Cold-call 4 learners to share one change. Then introduce the plan: 'Dr. Wangari Maathai started with seedlings in Kenya and the Green Belt Movement planted millions of trees. We have 60 seedlings ready.' Ask them to check the arithmetic, then probe: 'Why is KES 20 a fair price?' 'Why not sell for less than the KES 300 cost?' 'How does selling a seedling help conservation in Kitale?' 'What should we tell a buyer about watering, given what we know about oxygen and waterlogged seeds?' Ensure the plan includes a role for each member (Kiprop as nursery lead, record-keeper, cashier, communicator, customer care), supporting Unity. If the class is large, give each group a different task (pricing, advertising, care instructions, record keeping) and combine them. Finish with the exit ticket: 'What evidence did we gather today?' and 'How does this help answer our driving question?' Collect and read these for reflection. Close with: 'You are no longer just learners, you are Kitale's tree growers.'	model next to the final model gives learners concrete evidence of conceptual change and strengthens metacognition (Learning to Learn). Linking the model to the nursery plan shows that scientific understanding (oxygen, dormancy, hormones, rooting) guides real decisions about conservation and income.	model includes (1) seed structure difference (monocot vs dicot), (2) seed condition (8 dormant beans, non-viable seeds, pests), (3) environment (waterlogging, oxygen), (4) primary growth and apical dominance with hormones. Peer review of models using the ticks and stars. Profit calculation accuracy: KES 1,200, KES 300, KES 900. Exit ticket indicators: names at least two pieces of evidence and states how they answer the driving question. Observation checklist for Unity and Citizenship: all members contribute, roles are shared, and the plan explicitly benefits the community.
--	---	--	--	--

D. TEACHER REFLECTION

After the lesson, review: (1) How many learners moved from attributing differences to rain or planting date to giving a multi-cause explanation using data? (2) Which misconceptions persisted, such as maize having secondary growth, confusing dormancy with non-viability, or mixing germination and emergence? (3) Did every learner contribute during the gallery walk and group planning, or did a few dominate? (4) Were the time-lapse videos or the printed alternatives sufficient where there was no electricity? (5) Did learners accurately calculate $\text{KES } 1,200 - \text{KES } 300 = \text{KES } 900$ and link the plan to conservation? (6) What follow-up should be done for learners who need support, for example a short revision quiz on seed structure, dormancy, growth and hormones, and how will the nursery be sustained after this unit?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed our own Day 1 models compared with our final models, the evidence on other groups' posters (germination 88 maize vs 80 beans out of 100; 8 dormant beans, 7 germinating after scarification; emergence Day 7 for maize and Day 8 for beans; heights at Day 42 of 125 cm for maize and 44 cm for beans; 0 branches vs 3 branches after removing the tip), and the nursery seedling sales plan ($60 \times \text{KES } 20 = \text{KES } 1,200$, cost KES 300, profit KES 900).
What did I learn?	We learned that the differences were not caused by the rain or planting date. They come from seed structure (maize monocot and hypogeal; beans dicot and epigeal), seed condition (dormancy from a hard seed coat, non-viable seeds, pests) and environment (waterlogging and lack of oxygen), and that hormones (auxin, gibberellins, cytokinins, abscisic acid) coordinate primary growth and branching. We also learned that our science can be used to grow and sell tree seedlings for conservation and income, as Dr. Wangari Maathai inspired.
How does this explain the phenomenon?	Claim: The maize and beans on Mr. Wafula's farm differ because of seed structure, seed condition and environment, not because of rain or planting date. Evidence: Day 14 germination of 88 maize vs 80 beans; the failures were pests, non-viable embryos, waterlogging and, in beans, 8 dormant seeds, 7 of which germinated after scarification; maize emerged on Day 7 and beans on Day 8; Day 42 heights were 125 cm and 44 cm; beans stayed unbranched with the tip but formed 3 branches after the tip was removed. Reasoning: structure decides how seedlings emerge, a hard testa blocks water, no oxygen stops respiration, meristems and hormones coordinate growth, and apical dominance explains branching, so farmers can improve yields and communities can grow trees.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.